

Split support through the original arithmetic observation

Support, signed-current stability and longer arithmetic evolution

Split-Zero research programme

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The original split support and its mixed cubic maps survive the programme's actual minimum-section observation in quantitatively controlled forms. This continuation calculates both measured support eigenvalues, their complete hidden-kernel defect, and their effect on the terminal arithmetic class. A complete finite family of ordered mixed products recovers the original signed current with amplification less than six, uniformly in the original degree. Positive mixed products recover the measured support itself. The new deformation has a shifted collision whose imaginary displacement records the current trace. All original source metrics, period guards, complex phases and four cutoffs remain in the proofs.

The signed current has a finite metric-stability bound and an exact quadratic numerator along each original source step. Growing source-jet control extends the measured and intrinsic arithmetic evolution, including their actual inverses, to a longer time interval and evaluates its lower condition-number plateau. DF15–16 and EE47–48 correct the constant name in the earlier RD15: the response coefficient is $4[\psi(0) - \psi(1)] = 0.7001820930\dots$; the kernel-volume constant remains $C_{\mathcal{O}} = 1.3542819878\dots$. The exact integral map connecting the two is proved in those sections.

The new results strengthen the support receiver and terminal-current calculation. The particular terminal signed mixed-product value remains to be evaluated; no Riemann-hypothesis conclusion is asserted. Complete new proofs, exact auxiliary checks and the preceding proof collection accompany this reader. Pinned public sources and inherited human citations are given at their uses.

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MP7-24: ordered mixed support retains the signed current

$$X_g = \sqrt{\varepsilon} f g^\dagger$$

$$H = \{e, f\}^\perp$$



$$\text{span}\{e, f\}$$



$$Y_g = \sqrt{\varepsilon} g e^\dagger$$

$$X_g Y_g = R, \quad Y_g X_g = 0$$

Original observation: $F \mapsto J^\dagger F J$

$$\mathbb{E}(X_{B,g} Y_{B,g}) = s R_B$$

$$i \mathbb{E}(X_B Y_B - Y_B^* X_B^*) = s W_B, \quad 1/s < 6$$

ST8: both support eigenvalues, including the second direction

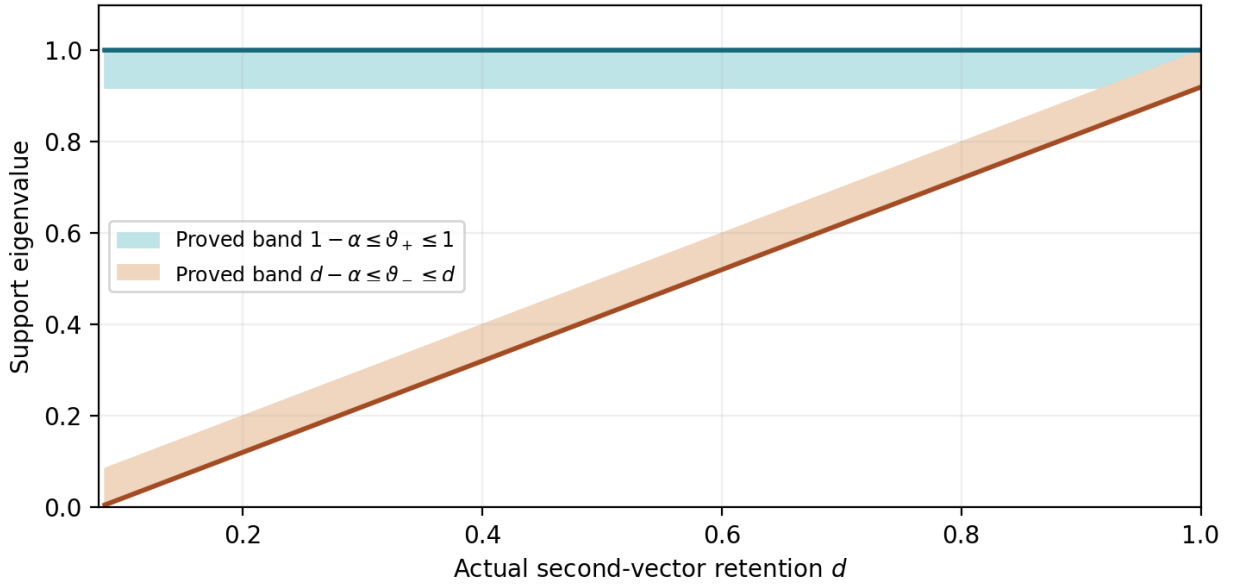


Figure 1: Upper: the two actual support blocks and their ordered product through the original isometry, proved in MP7-24. The average uses every vector of an orthonormal basis of the original complement; the result is independent of that basis. Its inverse factor is bounded by the original five-orbit rank theorem. Lower: the exact ST8 bands. To show both directions, the illustrative family uses $\alpha = 0.08$ and $|r|^2 = \alpha(1 - d)$, $d > \alpha$, giving exactly $\vartheta_+ = 1$, $\vartheta_- = d - \alpha$. It is a positive contraction family, not a numerical sample of native period data. Every actual matrix obeys the proved bands without that illustration's parameter choice.

DF3-13: the collision is shifted by the full kernel return

$$F_B(s) = (\varepsilon v + su)u^*, \quad \lambda(s) = as + \varepsilon r$$

$$\text{Native } s = 0 \\ R_B^2 = \varepsilon r R_B$$

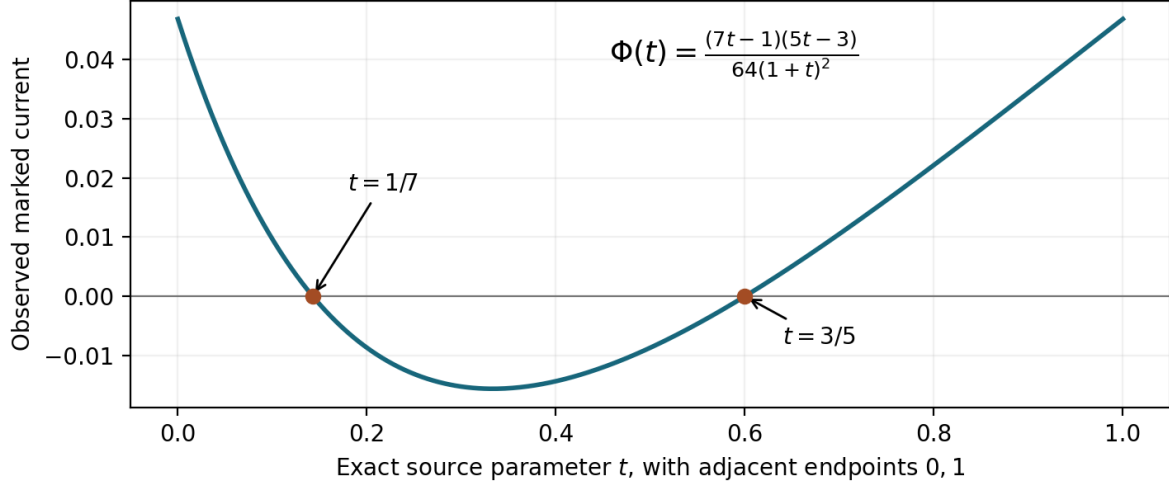


$$s_* = -\varepsilon r/a \\ \bar{R}^2 = 0$$

$$\text{Im } s_* = \text{tr } W_B(0)/(2a)$$

A real parameter reaches the collision exactly when the current trace vanishes.

SC18-23: two sign crossings with the same eigenclass



EE39: the complete original four-cutoff condition profile

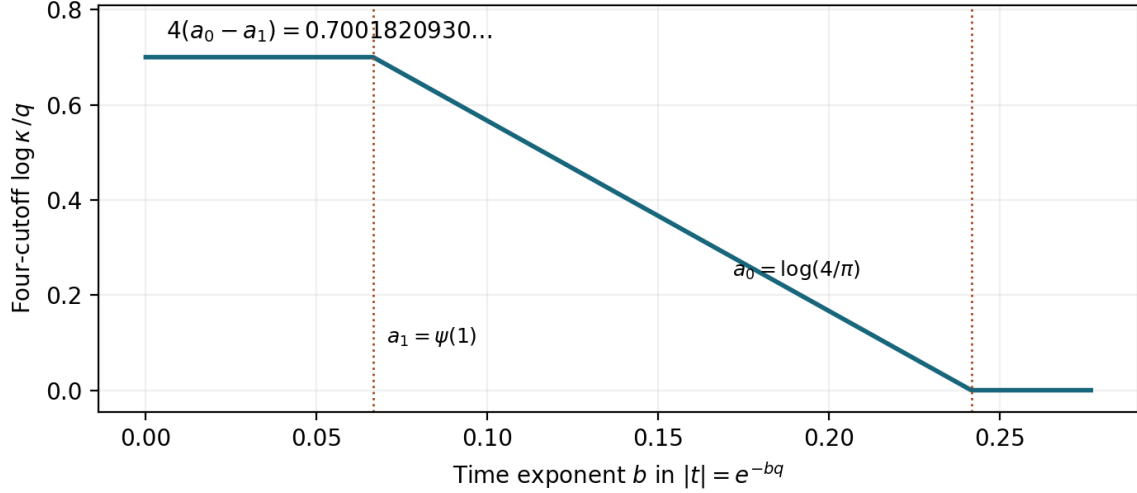


Figure 2: Upper: the exact native-to-collision parameter map DF3-13. The source family is the pinned ID1-24 construction; its original-metric receiver is proved below. Middle: the exact finite example of SC23, with its specified fixed M, x, K and complete positive metric $G(t) = (I + twu^*)^{-1}$. The two crossings $1/7, 3/5$ attain the quadratic degree bound of the original source-step formula. This is an auxiliary finite example, not a native-period sign calculation. The coefficient formulas and full metric are retained in the proof. Lower: EE39's limiting four-cutoff profile $4[(a_0 - b)_+ - (a_1 - b)_+]$, with its proved $O(k \log q)$ unscaled error. The values $a_i = \psi(i)$ come from the original elliptic formulas, EE47-48. The new lower plateau holds for each fixed $0 < b < a_1$ and at the proved subexponential clock; the picture does not assign a value at literal $t = 1$.

What the original observation retains of split support

22 September 2026. Result SZ-20260922-028, root derivation ST1–ST20.

The support idempotent has two measured eigenvalues. We calculate both, their exact defect through the original invariant kernel, and their effect on the original terminal class. One eigenvalue approaches one at the doubled exponential rate already proved for boundary leakage. The other differs from the actual retention of the second boundary vector by at most that same leakage. We then calculate how the measured support and its defect determine the terminal class's combined current-sector mass, with an absolute error at that rate. All statements below concern the original observation and its complete source metric.

1. Original objects, source provenance and domain

Keep the stipulated simple quartet, its original admitted period, the five-orbit observation, and

$$k \geq 17, \quad k \equiv 1 \pmod{4}, \quad q = (k+1)^2, \quad (ST1)$$

$$Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad E = \mathbb{C}[y]/Q_k.$$

At each literal source cutoff $q-1 \leq N \leq 2q$, retain the complete original source metric G_N , observation $\Lambda : E \rightarrow B$, and all of $K = \ker \Lambda$. A dagger denotes the adjoint in this metric. The original minimum section and its Euclidean-coordinate isometry are

$$Q_B = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L = G_N^{-1} \Lambda^* Q_B, \quad J = L Q_B^{-1/2}, \quad (ST2)$$

$$J^\dagger J = I_B, \quad P_B = J J^\dagger = L \Lambda.$$

The use of J preserves the metric: $Q_B^{1/2} \Lambda$ restricted to $K^\perp$ is its inverse. In particular none of the following operators changes the original minimum. These facts and the exact boundary decomposition

$$M = C + R, \quad C = C^\dagger, \quad R = \epsilon f e^\dagger, \quad e^\dagger f = 0, \quad \|e\| = \|f\| = 1, \quad (ST3)$$

$$\epsilon > 0$$

are proved in CE2–8. The vectors are the actual consecutive monic source classes $e = b_N/\sqrt{E_N}$, $f = b_{N+1}/\sqrt{F_N}$, and $\epsilon = \sqrt{E_N F_N}/\omega_N$.

Write $\Pi = ee^\dagger + ff^\dagger$, $H = \text{im}(I - \Pi)$. The support-algebra map is

$$\rho : \mathbb{C} \times \mathbb{C}[\eta]/(\eta^2) \longrightarrow \text{End}(E), \quad (ST4)$$

$$(c_0, c_1 + c_2 \eta) \longmapsto c_0(I - \Pi) + c_1 \Pi + c_2 R.$$

For completeness, $\Pi^2 = \Pi$, $\Pi R = R \Pi = R$, and $R^2 = 0$ prove multiplicativity. Restriction to $H \neq 0$ recovers c_0 , application to f recovers c_1 , and then application to e recovers c_2 ; thus the map is faithful. It receives the original additive-monoid algebra of $\mathbb{Z} \sqcup \{\tau\}$: $[\tau] \mapsto I$, $[n] \mapsto \Pi + nR$. The supported-zero symbol maps to Π ; multiplication of these symbols means addition in that original monoid. Its kernel is exactly the ideal $([1] - [0])^2$, since the monoid algebra is $\mathbb{C} \times \mathbb{C}[t^{\pm 1}]$, with $t \mapsto 1 + \eta$. This is the support input of NI19–29 in the pinned node edition; every part required below has just been proved here.

Define measured columns and support without replacing either source projection:

$$u = J^\dagger e, \quad v = J^\dagger f, \quad U = [u, v],$$

$$C = U^* U = \begin{pmatrix} a & r \\ \bar{r} & d \end{pmatrix}, \quad \mathfrak{d} = ad - |r|^2, \quad (ST5)$$

$$A = J^\dagger \Pi J = U U^*, \quad \alpha = 1 - a.$$

Here A denotes measured support throughout this note; it is distinct from the compressed energy denoted A_N in older determinant papers. On the proved finite area guards,

$$0 \prec C \preceq I_2, \quad \mathfrak{d} \geq \Theta_N > 0, \quad |r|^2 \leq \alpha(1 - d),$$

$$\alpha \leq e^{-2q\psi(t_N) + O_{h,\omega}(k \log(q+2))}, \quad -\log \Theta_N = O_{h,\omega}(k \log(q+2)), \quad (ST6)$$

$$\log \epsilon = q\psi(t_N) + O_{h,\omega}(k \log(q+2)), \quad t_N = (N+1-q)/q.$$

These are the original complete-source and conductor estimates RC25. The exact polynomial, Gamma and elliptic sources remain those cited there: T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, DLMF Chapter 18; R. A. Askey and R. Roy, Chapter 5; B. C. Carlson, Chapter 19. The finite linear algebra below is proved in full and needs no additional external theorem.

2. Both support eigenvalues and the full kernel defect

The two nonzero eigenvalues of A are exactly

$$\vartheta_{\pm} = \frac{a + d \pm \sqrt{(a - d)^2 + 4|r|^2}}{2}. \quad (\text{ST7})$$

Indeed if $Cz = \vartheta z$, then $Uz \neq 0$ and $A(Uz) = UCz = \vartheta Uz$. Conversely every nonzero eigenvector of A lies in $\text{im } U$; this proves the correspondence including multiplicities. The quadratic formula applied to $\det(\lambda I - C)$ proves ST7. All remaining eigenvalues of A are zero, of multiplicity $b - 2$, where $b = \dim B$.

There is a bound valid even when these eigenvalues collide:

$$\boxed{0 \leq 1 - \vartheta_+ \leq \alpha, \quad 0 \leq d - \vartheta_- \leq \alpha, \quad \vartheta_- \geq \Theta_N.} \quad (\text{ST8})$$

The Rayleigh quotient at the first coordinate gives $\vartheta_+ \geq a = 1 - \alpha$, while $C \preceq I$ gives $\vartheta_+ \leq 1$. Put $\ell = 1 - \vartheta_+$. Since $\vartheta_+ + \vartheta_- = a + d$, one has $\vartheta_- = d - \alpha + \ell$, giving the second assertion. Finally $\vartheta_- = \mathfrak{d}/\vartheta_+ \geq \mathfrak{d} \geq \Theta_N$. There is no division by an eigenvalue gap.

The full failure of multiplicativity is a positive kernel square:

$$\Phi(XY) - \Phi(X)\Phi(Y) = J^\dagger X(I - P_B)YJ, \quad \Phi(X) = J^\dagger XJ,$$

$$\boxed{A - A^2 = D^\dagger D, \quad D = (I - P_B)\Pi J.} \quad (\text{ST9})$$

Insert $P_B = JJ^\dagger$ to prove the first identity, then take $X = Y = \Pi$. Thus the defect includes precisely the original kernel projection. Its two possibly nonzero eigenvalues are $\vartheta_{\pm}(1 - \vartheta_{\pm})$, retaining the labels inherited from ST7 rather than reordering them. Since $|x(1 - x) - y(1 - y)| = |x - y||1 - x - y| \leq |x - y|$ on $[0, 1]$, ST8 gives

$$0 \leq \vartheta_+(1 - \vartheta_+) \leq \alpha, \quad |\vartheta_-(1 - \vartheta_-) - d(1 - d)| \leq \alpha. \quad (\text{ST10})$$

Equivalently the two corresponding singular values of D are $\sqrt{\vartheta_{\pm}(1 - \vartheta_{\pm})}$. The second differs from $\sqrt{d(1 - d)}$ by at most $\sqrt{\alpha}$, using $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$. The first is at most $\sqrt{\alpha}$. This retains the second mixed direction even when its defect is large.

The exact support volume has the stronger scalar receiver

$$\boxed{0 \leq \log d - \log \det^+ A \leq \frac{\alpha}{\Theta_N}.} \quad (\text{ST11})$$

Here $\det^+ A = \vartheta_+ \vartheta_- = \mathfrak{d}$. Write $d - \mathfrak{d} = \alpha d + |r|^2 \leq \alpha$ by ST6. Therefore $\log(d/\mathfrak{d}) = \log(1 + (d - \mathfrak{d})/\mathfrak{d}) \leq \alpha/\Theta_N$. This is a comparison with the actual $d = \|P_B f\|^2$, not an assumed unit retention.

At all four original cutoffs with signs $+, +, -, -$,

$$\begin{aligned} & |\mathcal{R} \log \det^+ A - \mathcal{R} \log d| \\ & \leq \sum_{N \in \{q-1, q, 2q-1, 2q\}} \frac{\alpha_N}{\Theta_N}. \end{aligned} \quad (\text{ST12})$$

Since $\Theta_N \leq d_N \leq 1$, the entire support-volume return is $O_{h, \omega}(k \log(q + 2)) = o(kq)$. The bound in ST12 is at the doubled exponential rate separately at the four arguments $0, 1/q, 1, 1 + 1/q$. No signed value is assigned to $\mathcal{R} \log d$.

3. The original terminal class retains measured support to doubled accuracy

Use the exact terminal cardinal $x = v_+$ of TR1, with $Mx = (k\gamma - ik\delta)x$. Keep precisely the established terminal domain $\gamma \geq 1000\delta$, the original five-orbit requirement and $|\varpi| \geq R_{\text{corner}}$, with the finite radius of RC37-42. This domain proves $P_B x \neq 0$. In the J -coordinates define

$$\begin{aligned}\hat{y} &= J^\dagger x / \|P_B x\|, & g &= u/\sqrt{a}, & h &= (v - (r/a)u)/\sqrt{\mathfrak{d}/a}, \\ c_g &= g^* \hat{y}, & c_h &= h^* \hat{y}.\end{aligned}\tag{ST13}$$

These are the exact observed orthonormal coordinates of TR9-11. The complete terminal equation proves $|c_g| \leq \eta_N \leq e^{-q\psi(t_N) + O_{h,\varpi}(k \log(q+2))}$, with finite η_N in TR10. Let

$$\begin{aligned}\Pi_{\text{range}} &= UC^{-1}U^*, & \mu &= \hat{y}^* \Pi_{\text{range}} \hat{y} = |c_g|^2 + |c_h|^2, \\ \mathfrak{s} &= \hat{y}^* A \hat{y}.\end{aligned}\tag{ST14}$$

Multiplication proves that Π_{range} is the orthogonal range projector. It is retained alongside the measured weights in A .

The exact measured support in this frame is

$$A|_{g,h} = \begin{pmatrix} a + |r|^2/a & r\sqrt{\mathfrak{d}/a} \\ \bar{r}\sqrt{\mathfrak{d}/a} & d - |r|^2/a \end{pmatrix}.\tag{ST15}$$

This follows by substituting $u = \sqrt{a}g$ and $v = (r/\sqrt{a})g + \sqrt{\mathfrak{d}/a}h$ in UU^* . All complex phases remain. Because $0 \preceq C \preceq I$, diagonalization gives $C^2 \preceq C$, whose first diagonal entry implies $|r|^2/a \leq 1 - a = \alpha$. Also $a + |r|^2/a \leq 1$ and $\mathfrak{d}/a \leq 1$. Thus

$$|\mathfrak{s} - d\mu| \leq (\eta_N + \sqrt{\alpha})^2 =: E_N^{\text{supp}}.\tag{ST16}$$

To prove the bound explicitly, subtract $d\mu$ from the quadratic form ST15. The coefficient of $|c_g|^2$ is $a + |r|^2/a - d$, of absolute value at most one. The other diagonal term has absolute value at most $\alpha|c_h|^2$. The off-diagonal term is bounded by $2\sqrt{\mathfrak{d}/a}\sqrt{|r|^2/a}|c_g||c_h| \leq 2\sqrt{\alpha}\eta_N$. Since $|c_h| \leq 1$, their sum proves ST16, with no lower bound on μ .

There is also a complete terminal defect estimate:

$$|\hat{y}^*(A - A^2)\hat{y} - d(1 - d)\mu| \leq 2E_N^{\text{supp}}.\tag{ST17}$$

Set $F = A - d\Pi_{\text{range}}$. Its eigenvalues on the current plane are $\vartheta_\pm - d$, each in $[-1, 1]$, so $\|F\| \leq 1$. Its second column in ST15 has squared norm $\mathfrak{d}|r|^2/a^2 + |r|^4/a^2 = d|r|^2/a \leq \alpha$. Hence $\|F\hat{y}\| \leq |c_g| + \sqrt{\alpha}|c_h| \leq \eta_N + \sqrt{\alpha}$. Now $A - A^2 - d(1 - d)\Pi_{\text{range}} = (1 - 2d)F - F^2$. Its expectation is bounded by ST16 plus $\|F\hat{y}\|^2$, proving ST17. Components orthogonal to the current plane vanish from all these operators, but remain in the unit norm of $\hat{y}$.

In particular the combined current-sector mass has the quantitative receiver

$$\left| \mu - \frac{\mathfrak{s}}{d} \right| \leq \frac{E_N^{\text{supp}}}{d} \leq \frac{E_N^{\text{supp}}}{\Theta_N} \leq e^{-2q\psi(t_N) + O_{h,\varpi}(k \log(q+2))}.\tag{ST18}$$

If z_+, z_- are the canonically phased current amplitudes TR14-18, then $\mu = |z_+|^2 + |z_-|^2$, and exactly

$$\mathfrak{s} - 2d\Re(z_+ \bar{z}_-) = (\mathfrak{s} - d\mu) + d|z_+ - z_-|^2.\tag{ST19}$$

Consequently its absolute value is at most $E_N^{\text{supp}} + d(\sqrt{2}\eta_N + \delta_N^{\text{tr}})^2$, also at the doubled rate. This carries support through to the real current coherence, including zero amplitude cases. It is stronger than an assertion that the two support presentations correspond: the complete terminal error has been evaluated in the original metrics.

4. Receiver into the signed current and scope

The exact signed current is

$$j_N = 2\epsilon\sqrt{\mathfrak{d}}\Im(\bar{c}_g c_h) - 2\epsilon(\Im r)|c_g|^2.$$

Its measured-support bound therefore improves the mass-free magnitude estimate to

$$\boxed{|j_N| \leq 2\epsilon\sqrt{\mathfrak{d}}\eta_N \quad \sqrt{\min\{1, (\mathfrak{s} + E_N^{\text{supp}})/d\}} + 2\epsilon|r|\eta_N^2.} \quad (\text{ST20})$$

Use $|c_h|^2 \leq \mu \leq \min\{1, (\mathfrak{s} + E_N^{\text{supp}})/d\}$ in the exact current expression. The last term remains the absolutely doubled-exponential error of TR20. This bound retains the actual measured support; it does not assume that its value is nonzero or replace it by one. The companion mixed-probe derivation calculates the ordered products carrying the imaginary part through the same observation. Together these are distinct quantitative receivers of the new support data: ST18 controls current-sector mass, while the ordered mixed products retain orientation. A sign for the particular terminal class still requires its signed mixed-product value. No RH conclusion is inferred from a positive support bound.

All four cutoffs remain separate. For any of ST16–19, the error of its signed four-cutoff return is bounded by the sum of the four displayed absolute errors; the proof is the triangle inequality with exactly the signs $+, +, -, -$. No cutoff or full-kernel term has been dropped.

Canonical mixed cubic probes recover the original support and signed current

22 September 2026. The node deformation supplies the complete family T_g whose square is the original rank-one arithmetic primitive. We calculate its finite averages after the original observation, including every hidden return. Keeping the two original support blocks in their order recovers the exact signed current with a uniformly bounded factor. Averaging positive second moments also recovers the actual measured support.

1. Original metrics, support blocks, and finite family

At every original cutoff $q-1 \leq N \leq 2q$, retain the full quotient (E, G_N) , $q = (k+1)^2$, $k \equiv 1 \pmod{4}$, $k \geq 17$, original quartet, period, branch and arithmetic unit. The exact action, from OCP5-7, is

$$M = C + R, \quad C = C^\dagger, \quad R = \epsilon f e^\dagger, \quad \epsilon > 0, \quad \|e\| = \|f\| = 1, \quad e^\dagger f = 0. \quad (\text{MP1})$$

Every source adjoint in this proof is the G_N -adjoint. The original onto map $\Lambda : E \rightarrow B$, its complete quotient metric and minimum section give

$$Q_B = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L = G_N^{-1} \Lambda^* Q_B, \quad J = L Q_B^{-1/2}, \quad J^\dagger J = I_B, \quad P = J J^\dagger = L \Lambda. \quad (\text{MP2})$$

Thus P is the actual orthogonal observation projection and $I - P$ projects onto the full original invariant kernel. The operator $\Phi(F) = J^\dagger F J$ is written in the isometric coordinates of the original Q_B metric. Its complete product defect is

$$\Phi(FG) - \Phi(F)\Phi(G) = J^\dagger F(I - P)GJ. \quad (\text{MP3})$$

This follows by inserting $J J^\dagger = P$; it does not assume that Φ preserves products.

Let

$$\Pi = e e^\dagger + f f^\dagger, \quad P_H = I - \Pi, \quad H = \{e, f\}^{\perp_{G_N}}, \quad d_H = \dim H = q - 2.$$

The original support and observed source columns are

$$u = J^\dagger e, \quad v = J^\dagger f, \quad A = J^\dagger \Pi J = u u^* + v v^*, \quad a = u^* u, \quad \beta = v^* v, \quad r = u^* v, \quad R_B = \Phi(R) = \epsilon v u^*. \quad (\text{MP4})$$

Keep A , a positive contraction, with its actual nonidempotence. Define

$$Z = \frac{J^\dagger P_H J}{d_H} = \frac{I_B - A}{d_H}, \quad s = \text{tr } Z \\ = \frac{\dim B - a - \beta}{d_H} = 1 - \frac{\text{tr}(P_H(I - P))}{d_H}. \quad (\text{MP5})$$

All traces are the metric traces of the stated finite spaces. In particular $0 \preceq Z \preceq I_B/d_H$ and $0 \leq s \leq 1$.

Take any complete orthonormal basis $g_1, \dots, g_{d_H}$ of the actual space H . The finite averaging operation used below is

$$\mathbb{E}F(g) = \frac{1}{4d_H} \sum_{j=1}^{d_H} \sum_{\lambda \in \{1, i, -1, -i\}} F(\lambda g_j). \quad (\text{MP6})$$

There is no distinguished basis vector or omitted direction. We prove that each displayed average is independent of the chosen orthonormal basis. The family is a finite exact realization of the required second moments, not a stochastic approximation.

For each unit $g \in H$, define the original mixed cubic operator and its two ordered support blocks:

$$T_g = \sqrt{\epsilon}(f g^\dagger + g e^\dagger), \quad X_g = \Pi T_g P_H = \sqrt{\epsilon} f g^\dagger, \quad Y_g = P_H T_g \Pi = \sqrt{\epsilon} g e^\dagger. \quad (\text{MP7})$$

Orthogonality gives

$$X_g Y_g = R, \quad Y_g X_g = 0, \quad X_g^2 = Y_g^2 = 0, \quad T_g^2 = R, \quad T_g^3 = 0. \quad (\text{MP8})$$

These are exactly the mixed support maps supplied by NI9-13/SO9. The order of X_g, Y_g matters before and after observation.

2. All quadratic averages, with phases and hidden terms retained

Put $w_g = J^\dagger g$, so

$$\begin{aligned} T_{B,g} &= \Phi(T_g) = \sqrt{\epsilon}(vw_g^* + w_g u^*), & X_{B,g} &= \sqrt{\epsilon}vw_g^*, \\ Y_{B,g} &= \sqrt{\epsilon}w_g u^*. \end{aligned} \quad (\text{MP9})$$

For $g = \lambda g_j$, $w_g = \lambda w_j$; hence

$$\begin{aligned} \mathbb{E}(w_g w_g^*) &= Z, & \mathbb{E}\|w_g\|^2 &= s, & \frac{1}{4} \sum_{\lambda} \lambda^2 \\ &= \frac{1}{4} \sum_{\lambda} \bar{\lambda}^2 = 0. \end{aligned} \quad (\text{MP10})$$

The first equality uses $\sum_j g_j g_j^\dagger = P_H$. This proves basis independence, since the final expression depends only on the original projection P_H .

Multiplying MP9, the terms with phase $\lambda^{\pm 2}$ cancel, and the remaining exact averages are

$$\boxed{\mathbb{E} T_{B,g}^2 = sR_B + \epsilon r Z}, \quad (\text{MP11})$$

$$\boxed{\begin{aligned} \mathbb{E}(T_{B,g}^* T_{B,g}) &= \epsilon(\beta Z + s u u^*), & \mathbb{E}(T_{B,g} T_{B,g}^*) \\ &= \epsilon(a Z + s v v^*). \end{aligned}} \quad (\text{MP12})$$

For example, the four terms in $T_{B,\lambda g_j}^2/\epsilon$ are

$$\bar{\lambda}^2 v(w_j^* v)w_j^* + \|w_j\|^2 v u^* + r w_j w_j^* + \lambda^2 w_j(u^* w_j)u^*.$$

This verifies that the coefficient of Z in MP11 is $r = u^* v$, not its conjugate. The adjoint square average is therefore $\mathbb{E}(T_{B,g}^*)^2 = sR_B^* + \epsilon \bar{r} Z$. The same multiplication gives MP12: its omitted cross terms carry exactly $\lambda^{\pm 2}$.

On the full source, without observation,

$$T_g^\dagger T_g = \epsilon(gg^\dagger + ee^\dagger), \quad T_g T_g^\dagger = \epsilon(ff^\dagger + gg^\dagger).$$

Using MP3 and MP10-12 gives the complete hidden returns

$$\boxed{\mathbb{E} J^\dagger T_g(I - P)T_g J = (1 - s)R_B - \epsilon r Z}, \quad (\text{MP13})$$

$$\boxed{\begin{aligned} \mathbb{E} J^\dagger T_g^\dagger(I - P)T_g J &= \epsilon[(1 - \beta)Z + (1 - s)u u^*], \\ \mathbb{E} J^\dagger T_g(I - P)T_g^\dagger J &= \epsilon[(1 - a)Z + (1 - s)v v^*]. \end{aligned}} \quad (\text{MP14})$$

Both expressions in MP14 are positive semidefinite, as are their unaveraged complete squares. The adjoint of MP13 is the corresponding adjoint-square hidden return. The full original kernel appears in every $I - P$; no conductor-only kernel has been substituted.

The ordered products require no phase averaging at all: each pair of phases cancels within the product. The complete basis average, equivalently MP6, gives

$$\boxed{\mathbb{E}(X_{B,g} Y_{B,g}) = sR_B, \quad \mathbb{E}(Y_{B,g} X_{B,g}) = \epsilon r Z}. \quad (\text{MP15})$$

Indeed $X_{B,g} Y_{B,g} = \|w_g\|^2 R_B$, whereas $Y_{B,g} X_{B,g} = \epsilon r w_g w_g^*$. The corresponding exact hidden products are

$$\boxed{\mathbb{E} J^\dagger X_g(I - P)Y_g J = (1 - s)R_B, \quad \mathbb{E} J^\dagger Y_g(I - P)X_g J = -\epsilon r Z}. \quad (\text{MP16})$$

Thus the reverse product, which is zero on the source, is exactly cancelled by its actual hidden return after observation.

The four positive ordered hidden returns are likewise

$$\begin{aligned} \mathbb{E} J^\dagger X_g^\dagger(I - P)X_g J &= \epsilon(1 - \beta)Z, \\ \mathbb{E} J^\dagger Y_g^\dagger(I - P)Y_g J &= \epsilon(1 - s)u u^*, \\ \mathbb{E} J^\dagger X_g(I - P)X_g^\dagger J &= \epsilon(1 - s)v v^*, \\ \mathbb{E} J^\dagger Y_g(I - P)Y_g^\dagger J &= \epsilon(1 - a)Z. \end{aligned} \quad (\text{MP17})$$

This is proved either by direct substitution of MP7 or by subtracting the respective observed positive product from its complete source product. Adding the first two and last two gives exactly MP14.

3. Exact isolation of the original signed current

The original centered arithmetic current is

$$W_B = J^\dagger i(M - M^\dagger)J = i(R_B - R_B^*), \quad \text{tr } W_B = -2\epsilon\Im r. \quad (\text{MP18})$$

For every original observed vector b , its physical current is the quadratic form of W_B on $Q_B^{1/2}b$; the central scalar $kI/2$ is removed by the same centered-current definition in OCP8. It has not been inserted or changed here.

The Hermitian ordered mixed probe is

$$\mathcal{C}_{\text{ord}} = i \mathbb{E}(X_{B,g}Y_{B,g} - Y_{B,g}^*X_{B,g}^*) = sW_B. \quad (\text{MP19})$$

When $s > 0$, this recovers the entire original current, including every marked-class pairing:

$$\boxed{W_B = s^{-1}\mathcal{C}_{\text{ord}}.} \quad (\text{MP20})$$

For every Hermitian perturbation E , this inverse has exactly the operator-norm error $\|s^{-1}E\| = \|E\|/s$. Thus its amplification is exactly $1/s$; it is not an unspecified conditioning factor. For a particular observed vector $\hat{b}$, its current is exactly $\hat{b}^*\mathcal{C}_{\text{ord}}\hat{b}/s$. This identifies the actual probe whose numerical or analytic value would evaluate the residual sign, without assigning that unknown value.

The unordered square average also has an exact current receiver. Put

$$\mathcal{C}_{\text{sq}} = i \mathbb{E}[T_{B,g}^2 - (T_{B,g}^*)^2].$$

Then MP11 gives

$$\begin{aligned} \mathcal{C}_{\text{sq}} &= sW_B + (\text{tr } W_B)Z, \\ \text{tr } \mathcal{C}_{\text{sq}} &= 2s \text{tr } W_B, \end{aligned}$$

$$\boxed{W_B = \frac{\mathcal{C}_{\text{sq}}}{s} - \frac{\text{tr } \mathcal{C}_{\text{sq}}}{2s^2}Z.} \quad (\text{MP21})$$

The averaged hidden signed return is $(1-s)W_B - (\text{tr } W_B)Z$; adding it to the observed expression gives the complete W_B , with no remaining trace term.

For $n = \dim B \geq 2$, the exact induced operator norm of the Hermitian inverse map in MP21 is

$$\boxed{\begin{aligned} \left\| E \mapsto \frac{E}{s} - \frac{\text{tr } E}{2s^2}Z \right\| \\ \text{Herm, op} \rightarrow \text{Herm, op} &= \frac{1}{s} + \frac{(n-2)\|Z\|}{2s^2} \leq \frac{3}{2s}. \end{aligned}} \quad (\text{MP22})$$

To prove this, fix a unit vector b , set $\zeta = b^*Zb$, and pair the output with b . The resulting real linear functional on Hermitian E is trace pairing with

$$F_b = s^{-1}bb^* - \frac{\zeta}{2s^2}I_B.$$

Since $0 \leq \zeta \leq \text{tr } Z = s$, its eigenvalue on b is positive and its other $n-1$ eigenvalues are nonpositive. Its trace norm is exactly $1/s + (n-2)\zeta/(2s^2)$. The supremum over $\|E\| \leq 1$ is that trace norm, attained by the Hermitian sign of F_b . Taking b in the top eigenspace of Z proves equality in MP22. Finally $\|Z\| \leq 1/d_H$ and $s = (n - \text{tr } A)/d_H \geq (n-2)/d_H$, since $\text{tr } A \leq 2$. These give its final upper bound. All signs and trace corrections in this inverse are therefore quantitatively controlled.

4. The original rank theorem makes the recovery factor uniformly bounded

The original simple-quartet five-orbit observation has the proved bound

$$\begin{aligned} n = \text{rank } \Lambda &\geq \\ r_k^- &:= \left\lfloor \frac{(k+1)^2}{5} \right\rfloor + \mathbf{1}_{5|k}. \end{aligned} \quad (\text{MP23})$$

Here the observation, period-dependent transformed quadric and rank are exactly those of the source OCS25-32. The relevant finite argument is reproduced to identify what is being used.

In the source's exact four Fourier coordinates, with weights 1, 2, 3, 4 modulo five, let $\mathcal{R}_k$ be the degree- k polynomial space and $\mathcal{I}_k$ its weight-zero span. The original transformed quadric is the nonzero $Q(x) = Q_0(H^{-1}x)$, where $Q_0 = t_{++}t_{--} - t_{+-}t_{-+}$ and H is the actual invertible period-and-unit matrix. Homogeneous restriction to the original exponential curve has kernel exactly $Q\mathcal{R}_{k-2}$. The proof groups monomials by their two counts (a, b) : their $q = (k+1)^2$ exponents are distinct because $\delta, \gamma > 0$, so their exponential functions are independent by the Vandermonde determinant. Within a fixed pair, monomial differences are divisible by Q_0 using $A^j - B^j = (A - B) \sum_{\ell=0}^{j-1} A^{j-1-\ell} B^\ell$. Transforming by the same H proves the stated kernel. Thus

$$n = \dim \mathcal{I}_k - \dim(\mathcal{I}_k \cap Q\mathcal{R}_{k-2}).$$

Fix a multiplicative monomial order and the actual leading monomial x^μ of Q . A row-echelon basis of the intersection has distinct weight-zero leading monomials, each divisible by x^μ . Its dimension is at most the number of degree- $k-2$ monomials of weight $-\text{wt } \mu$. This retains the actual quadric rather than choosing generic coefficients.

For an explicit count, let $D_j = \binom{j+3}{3}$ and $\varepsilon_j = 1$ for $j \equiv 0 \pmod{5}$, -1 for $j \equiv 1 \pmod{5}$, and zero otherwise. The root-of-unity filter gives

$$d_{5,j,0} = (D_j + 4\varepsilon_j)/5, \quad d_{5,j,r} = (D_j - \varepsilon_j)/5 \quad (r \neq 0).$$

Indeed for every nontrivial fifth root ζ^ℓ , $\prod_{a=1}^4 (1 - \zeta^{\ell a} z)^{-1} = (1 - z)/(1 - z^5)$; its degree- j coefficient is exactly ε_j . Substituting in the leading-monomial bound gives $n \geq d_{5,k,0} - \max_r d_{5,k-2,r}$. Since $D_k - D_{k-2} = (k+1)^2$, the five residues of $k \pmod{5}$ give precisely MP23. This reproduces the finite count in the cited original map, without a genericity assumption.

Combining MP5 and MP23 gives, at every original cutoff,

$$\boxed{s \geq \frac{\lfloor q/5 \rfloor + \mathbf{1}_{5|k} - 2}{q - 2} > 0, \quad \frac{1}{s} \leq \frac{q - 2}{\lfloor q/5 \rfloor + \mathbf{1}_{5|k} - 2} \leq \frac{1610}{309} < 6.} \quad (\text{MP24})$$

For the last uniform bound use $q \geq 324$ and $\lfloor q/5 \rfloor - 2 \geq q/5 - 3$. The function $5(q-2)/(q-15)$ decreases for $q > 15$, and its value at 324 is $1610/309$. Thus the ordered current recovery is uniformly bounded; the unordered inverse MP22 is at most $805/103 < 9$.

The exact obstruction to these receivers is also specified: $s = 0$ if and only if $P_H J = 0$, equivalently $\text{im } J \subseteq \text{span}\{e, f\}$. This follows from $d_H s = \|P_H J\|_{\text{HS}}^2$. In that case every w_g , and hence every observed mixed cubic probe, vanishes; the current in the supported two-plane can still be nonzero. The proved original rank bound excludes precisely this obstruction, because $n \geq 64 > 2$.

5. Positive mixed probes recover the actual measured support

Let $t = \text{tr } A = a + \beta$ and

$$\mathcal{P}_2 = \mathbb{E}(T_{B,g}^* T_{B,g} + T_{B,g} T_{B,g}^*).$$

The two positive moments in MP12 give the exact polynomial receiver

$$\boxed{\mathcal{P}_2 = \frac{\epsilon}{d_H} [tI_B + (n - 2t)A], \quad \text{tr } \mathcal{P}_2 = \frac{2\epsilon}{d_H} t(n - t).} \quad (\text{MP25})$$

To verify it, add MP12 to obtain $\epsilon(tZ + sA)$, then substitute $Z = (I - A)/d_H$ and $s = (n - t)/d_H$. Here $0 \leq t \leq 2$, whereas the original $n \geq 64$, so $n - 2t > 0$. Consequently

$$\boxed{t = \frac{n - \sqrt{n^2 - 2d_H \text{tr } \mathcal{P}_2 / \epsilon}}{2}, \quad A = \frac{d_H \mathcal{P}_2 / \epsilon - tI_B}{n - 2t}.} \quad (\text{MP26})$$

The displayed square-root branch is forced by $0 \leq t \leq 2 < n/2$; the other quadratic root is outside this interval. These are recoveries of A , not replacements of it by an idempotent. Its full original defect is still

$$A - A^2 = J^\dagger \Pi(I - P)\Pi J,$$

whose spectrum and terminal-class consequences are calculated by the companion support proof.

There is a finite stability bound on this exact positive-data image. Write $\mathcal{D} = (d_H/\epsilon)\mathcal{P}_2 = tI + (n - 2t)A$. For any two actual admissible support contractions A_0, A_1 of rank at most two, set $\eta = \|\mathcal{D}_1 - \mathcal{D}_0\|$. Their traces give

$$\begin{aligned} |t_1 - t_0| &\leq \frac{n\eta}{2(n-4)}, \\ \boxed{\|A_1 - A_0\| &\leq \frac{\eta}{n-4} \left(1 + \frac{n}{2(n-4)}\right)}. \end{aligned} \tag{MP27}$$

Indeed the trace difference is $2(n - t_1 - t_0)(t_1 - t_0)$, and its modulus is at most $n\eta$. Also

$$(n - 2t_1)(A_1 - A_0) = \mathcal{D}_1 - \mathcal{D}_0 - (t_1 - t_0)(I - 2A_0).$$

Since $0 \leq A_0 \leq I$, $\|I - 2A_0\| \leq 1$; both denominators are at least $n - 4 > 0$. This proves MP27. The estimate describes the inverse on its actual spectral image and does not assign an admissible support to arbitrary corrupted data.

6. Exact programme consequence and retained signed target

Equations MP19-24 show that the complete finite mixed family computes the original signed-current operator with a bounded loss. For every prescribed original terminal vector x , put $b = Q_B^{1/2}\Lambda x$. The same finite average gives

$$\boxed{\Phi_N(\Lambda x) = \frac{i}{s} \mathbb{E} [b^* X_{B,g} Y_{B,g} b - b^* Y_{B,g}^* X_{B,g}^* b]}. \tag{MP28}$$

The identity holds before any asymptotic expansion and at all four original cutoffs separately. It preserves the complete source minimum and every hidden return. The bounded factor does not evaluate an unevaluated complex product: the residual arithmetic sign still requires the value of the ordered quadratic average on the actual terminal vector. In particular relative errors measured against the large scale ϵ need not be small on the terminal current's smaller scale. MP20 quantifies the inverse error in absolute operator norm, so that distinction is explicit.

The node data have therefore supplied canonical finite probes for the actual original support and current, rather than an arbitrary distinguished square root. The averaged reverse product and every complementary return are part of the exact calculation.

Sources and proof locators

- Original source action and its complete minimum: OCP5-7.
- Actual metric isometry and product receiver: CE2-3, with the full hidden rank-one return at CE24-26.
- Actual period quadric and five-orbit rank: OCS28-32, with OCS31-32.
- Complete original node representation and mixed cubic family: NI8-13 in NATIVE_DUAL_NUMBER_RECEIVER.md, pinned node edition. The same edition retains the full support-algebra continuation NI17-29 and its ID1-24 idempotent-deformation companion. The mixed block formulas used here are explicitly rederived in MP7-8 and MP15-17.

The finite matrix, trace and root-of-unity calculations used here are proved in full. No additional external theorem is used. Human provenance for the inherited original-source construction remains in the cited complete programme proofs, including the DLMF Gamma and orthogonal-polynomial authors; these new finite averages do not alter those source attributions.

The support deformation after the original observation

22 September 2026. Result SZ-20260922-028, DF1-DF16.

The new idempotent deformation has an exact receiver through the original minimum section. Its collision point moves by a computable original-kernel return. The imaginary part of that displacement records the trace of the signed current. The real part moves the eigenvalues while preserving the full current. We calculate the resulting algebra, eigenline defect and metric cost, keeping the full original arithmetic operator distinguished from its boundary term.

1. The original family and its complete compression

Use all original objects, metrics and area guards of ST1–ST6. Let $P_e = ee^\dagger$. The new source construction is

$$F(s) = R + sP_e, \quad M(s) = C + F(s), \quad s \in \mathbb{C}. \quad (\text{DF1})$$

It is the family of ID1–ID24. The source arithmetic operator is exactly $M(0)$. The derivation of its receiver below does not identify other parameters with arithmetic multiplication. Since $R^2 = 0$, $P_e R = 0$, $R P_e = R$, and $P_e^2 = P_e$, one has $F(s)^2 = sF(s)$.

Compression uses the unchanged original J :

$$F_B(s) = J^\dagger F(s) J = (\epsilon v + su)u^*, \quad M_B(s) = J^\dagger M(s) J = C_B + F_B(s), \quad C_B = C_B^*. \quad (\text{DF2})$$

Set $\lambda(s) = as + \epsilon r$. Multiplication proves

$$\boxed{F_B(s)^2 = \lambda(s)F_B(s), \quad J^\dagger F(s)(I - P_B)F(s)J = (\alpha s - \epsilon r)F_B(s).} \quad (\text{DF3})$$

The first identity uses $u^*(\epsilon v + su) = \epsilon r + sa$; the second is the complete multiplicativity defect ST9 together with $F^2 = sF$. Thus both the supported and hidden terms are retained. The exact original current frame gives

$$F_B(s)|_{g,h} = \begin{pmatrix} \lambda(s) & 0 \\ \epsilon\sqrt{\mathfrak{d}} & 0 \end{pmatrix}. \quad (\text{DF4})$$

It vanishes on the orthogonal complement of this plane. All entries follow from ST13 and $\epsilon v + su = (\lambda(s)/\sqrt{a})g + \epsilon\sqrt{\mathfrak{d}/a}h$.

2. The exact family of algebras and its eigenline defect

Over $\mathbb{C}[s]$, the full unital observed algebra is faithfully

$$\begin{aligned} \mathbb{C}[s, z]/(z(z - \lambda(s))) &\longrightarrow \text{End}(B) \otimes \mathbb{C}[s], \\ z &\mapsto F_B(s), \quad 1 \mapsto I_B. \end{aligned} \quad (\text{DF5})$$

Monic division leaves a unique remainder $A(s) + B(s)z$. Its operator value on h gives $A(s) = 0$, and the h -component of its value on g gives $\epsilon\sqrt{\mathfrak{d}}B(s) = 0$, hence $B(s) = 0$. This proves the exact kernel and freeness of rank two over the base. The supported representation uses Π_{range} as unit and the same proof. The original measured support remains the different operator $A = J^\dagger \Pi J$; its full defect and weights are ST5–ST19.

The map of parameter lines $s \mapsto t = as + \epsilon r$ is an affine isomorphism, with inverse $s = (t - \epsilon r)/a$, because $a \geq \Theta_N > 0$. Thus DF5 is exactly the pullback of the source family $\mathbb{C}[t, z]/z(z - t)$ by this stated map. It is not the pullback by the identity parameter map unless its explicitly computed coefficients agree. At the literal native parameter $s = 0$, the observed fibre is $\mathbb{C}[z]/z(z - \epsilon r)$. Its nilpotent collision is instead at

$$\boxed{s_* = -\epsilon r/a.} \quad (\text{DF6})$$

At this point,

$$\begin{aligned} \tilde{R} = F_B(s_*) &= R_B - \frac{\epsilon r}{a}uu^* = \epsilon\sqrt{\mathfrak{d}}hg^*, \\ \tilde{R}^2 &= 0, \quad \tilde{R} \neq 0. \end{aligned} \quad (\text{DF7})$$

In particular its dual-number representation is faithful and its active amplitude is the exact observed area amplitude $\epsilon\sqrt{\mathfrak{d}}$.

The algebraic defect is also explicit before specializing. Write $b_0 = \epsilon\sqrt{\mathfrak{d}}$, a nonzero constant for the fixed original data. On the plane, the eigenline modules of DF4 are $\mathbb{C}[s]h$ and $\mathbb{C}[s](\lambda(s)g + b_0h)$. Their sum has cokernel

$$\boxed{\begin{aligned} &(\mathbb{C}[s]g \oplus \mathbb{C}[s]h) \\ &/(\mathbb{C}[s]h + \mathbb{C}[s](\lambda(s)g + b_0h)) \cong \mathbb{C}[s]/(s - s_*). \end{aligned}} \quad (\text{DF8})$$

The quotient sends $Ag + Bh$ to $A \bmod \lambda(s)$. Its kernel is exactly the displayed sum; surjectivity is witnessed by g . Since $\lambda(s) = a(s - s_*)$ with $a \neq 0$, the quotient is the stated one. This is the space defined by the collision obstruction, transported through the original observation. At s_* the two eigenlines meet in $\mathbb{C}h$, while the nonzero Jordan map DF7 remains.

3. Exact metric cost and the current carried by the displacement

For $s \neq s_*$, the two supported spectral idempotents are $E_1(s) = F_B(s)/\lambda(s)$, $E_0(s) = \Pi_{\text{range}} - E_1(s)$. Multiplication using DF3 proves all idempotent identities. Their exact norms are

$$\boxed{\begin{aligned} \|E_1(s)\| = \|E_0(s)\| &= \sqrt{1 + \frac{\epsilon^2 \mathfrak{d}}{|as + \epsilon r|^2}}, \\ \|F_B(s) - F_B(s_*)\| &= a|s - s_*|. \end{aligned}} \quad (\text{DF9})$$

In the orthonormal frame DF4, E_1 has first column $(1, b_0/\lambda)^T$ and second column zero. The only nonzero row of E_0 is $(-b_0/\lambda, 1)$. This proves both norms. The last identity follows from $F_B(s) - F_B(s_*) = (s - s_*)uu^*$, whose norm is $a|s - s_*|$. This is a metric calculation at the shifted original collision, including $\mathfrak{d}$ and a .

The complete observed arithmetic current of the deformed full operator is

$$W_B(s) = i(M_B(s) - M_B(s)^*) = W_B(0) - 2\Im(s)uu^*,$$

$$W_B(s)|_{g,h} = \begin{pmatrix} -2\Im\lambda(s) & -i\epsilon\sqrt{\mathfrak{d}} \\ i\epsilon\sqrt{\mathfrak{d}} & 0 \end{pmatrix}. \quad (\text{DF10})$$

The Hermitian C_B cancels exactly. Its two active eigenvalues are

$$\begin{aligned} &-\Im\lambda(s) \pm \sqrt{(\Im\lambda(s))^2 + \epsilon^2 \mathfrak{d}}, \\ &\det(W_B(s)|_{g,h}) = -\epsilon^2 \mathfrak{d}. \end{aligned} \quad (\text{DF11})$$

Expanding the characteristic polynomial proves both assertions. Thus the determinant of the active current is independent of the entire complex deformation, while its trace is $-2\Im\lambda(s)$. For real s , the whole current is unchanged, not only its determinant.

The exact obstruction to a real collision is

$$\boxed{\Im s_* = -\frac{\epsilon \Im r}{a} = \frac{\text{tr } W_B(0)}{2a}}. \quad (\text{DF12})$$

Consequently a real s reaches the collision if and only if the actual current trace is zero. The nearest real parameter is $s_{\mathbb{R}} = -\epsilon \Re r/a$. For every real s ,

$$|\lambda(s)|^2 = a^2(s - s_{\mathbb{R}})^2 + \epsilon^2(\Im r)^2.$$

If $\Im r \neq 0$, the maximum over the real line of the idempotent norm DF9 is therefore exactly $\sqrt{1 + \mathfrak{d}/(\Im r)^2}$, attained at $s_{\mathbb{R}}$. If $\Im r = 0$, the real path reaches the nonzero nilpotent fibre and the idempotent norms diverge as its parameter approaches that point. Both cases preserve the original measured columns. The obstruction defines this precise shifted family rather than ending the calculation.

At the complex collision, the relation to the original current is

$$W_B(s_*) = i(\tilde{R} - \tilde{R}^*), \quad W_B(s_*) - W_B(0) = 2\epsilon(\Im r)gg^*,$$

$$\begin{aligned} \frac{\|W_B(s_*) - W_B(0)\|}{\epsilon\sqrt{\mathfrak{d}}} &\leq 2\sqrt{\alpha/\Theta_N}, \\ |\hat{y}^*[W_B(s_*) - W_B(0)]\hat{y}]| &\leq 2\epsilon|r|\eta_N^2. \end{aligned} \quad (\text{DF13})$$

The first line follows from DF7 and DF10; the norm of gg^* is one. ST6 gives the relative bound, and ST13 gives the terminal bound. Thus the deformation identifies exactly the missing off-diagonal current of TR20 as the collision-fibre current. The terminal difference is absolutely $e^{-2q\psi(t_N)+O(k\log(q+2))}$. The collision fibre is tied to the native fibre by a proved small terminal correction, not by a declaration of equality.

4. Original scale and four-cutoff receiver

The shifted collision retains the exact estimate

$$\begin{aligned}\log \|\tilde{R}_N\| &= \log \epsilon_N + \frac{1}{2} \log \mathfrak{d}_N \\ &= q\psi(t_N) + O_{h,\varpi}(k\log(q+2)),\end{aligned}$$

$$\boxed{\mathcal{R} \log \|\tilde{R}_N\| = 2[\psi(0) - \psi(1)]q + O_{h,\varpi}(k\log(q+2)).} \quad (\text{DF14})$$

Here the return uses exactly $N = q-1, q, 2q-1, 2q$ with signs $+, +, -, -$, and the two monic endpoints are kept explicitly. ST6 bounds the logarithmic area correction; the endpoint modulus of the original elliptic profile, proved in RC8-10, gives the stated $O(\log q)$ changes under the two shifts by $1/q$. Moreover $|s_*| \leq \epsilon\sqrt{\alpha}/\Theta_N = e^{O(k\log(q+2))}$, while the active nilpotent amplitude grows at $e^{q\psi+O(k\log q)}$. These are scale statements with the original parameters retained. They do not set the collision displacement or its imaginary part to zero.

ST18 and the companion finite mixed-probe proof carry the same support and orientation data through the full original observation. DF13 identifies the residual signed terminal pairing with a precise point of the new algebraic family, up to its proved negligible correction. Evaluating its particular sign still requires that original marked pairing. The present result supplies the transported family, its exact obstruction module, metric cost, current determinant and four-cutoff amplitude; it does not infer spectral reality for $C_B + F_B(s)$ from the boundary algebra alone. The full C_B remains in DF2.

5. Correction propagated to the earlier denominator return

The new arrival identifies a constant-name error in the published027 RD15. Put $a_0 = \psi(0)$, $a_1 = \psi(1)$. The established kernel-volume constant is $C_\partial = 2J(1)$, whereas the boundary-amplitude return in DF14 has coefficient $2(a_0 - a_1)$. The exact relation, obtained by integrating the already proved RC10 and using $J(0) = 0$, is

$$J(t) = 4 \int_0^t \psi(s) ds - 2(1+t)\psi(t) + 2a_0,$$

$$\boxed{C_\partial = 8 \int_0^1 \psi(s) ds + 4a_0 - 8a_1.} \quad (\text{DF15})$$

Thus the full profile, with its integral retained, connects the two coefficients. The established values are $C_\partial = 1.35428198782529213288\dots$, $2(a_0 - a_1) = 0.350091046500433050\dots$. The earlier interval certificate for the first remains in024 CK14 and025 IR6; the second is evaluated by the same original RC7-10 elliptic formulas.

The original denominator estimate RD14 has $\log \mathfrak{A}_N = 2q\psi(t_N) + O(k\log q)$. Substituting all four literal arguments $0, 1/q, 1, 1+1/q$, with the endpoint bounds proved in RC8-10, gives the corrected return

$$\boxed{\mathcal{R} \log \mathfrak{A}_N = 4(a_0 - a_1)q + O(k\log q).} \quad (\text{DF16})$$

Its numerical coefficient is $0.700182093000866100\dots$. RD15's expression $2C_{\partial q}$ is withdrawn under the established kernel-volume notation. The separate denominator rates, their finite inequalities and all TR coherence estimates remain unchanged. The kernel return remains $mC_{\partial q} + o(kq)$. The preceding sealed edition is preserved as provenance; this successor supplies the corrected active equation and its exact connection to the older profiles.

Stable hidden-action calculation and exact source-step signs

22 September 2026. SC1–SC25. This rederives the newly supplied stable-sign continuation and carries it through the original mixed-probe receiver. The full rectangular action on the hidden kernel suffices to evaluate all three marked currents. Coherent changes of the complete metric have an error bound governed by this restricted action; the exponentially large current eigensectors need not be subtracted. The two original adjacent-cutoff paths give exact quadratic and quartic sign laws, with all coefficients retained.

1. Original full hidden action and the three marked currents

Keep ST1–ST3, the original fixed M, Λ, x , and $Mx = \omega x$. At an original metric G , write $h = P_K x$, $y = P_B x$, $Y = \|y\|_G^2$, $r_K = \|h\|_G / \|y\|_G$. The established terminal domain gives $Y > 0$. Every adjoint is in the stated original metric, conjugate-linear in its first argument. Define

$$\zeta = \langle y, Mh \rangle_G / Y, \quad \chi = \langle h, Mh \rangle_G / Y, \quad \xi = \omega r_K^2 - \chi. \quad (\text{SC1})$$

The eigenvalue equation and $h \perp_G y$ give $\langle y, My \rangle = \omega Y - \langle y, Mh \rangle$, and $\langle h, My \rangle = \omega \|h\|^2 - \langle h, Mh \rangle$. Hence

$$j_G(x) := \Phi_G(\Lambda x) / Y = -2\Im \omega + 2\Im \zeta. \quad (\text{SC2})$$

For $\omega = k\gamma - ik\delta$, its sign is exactly the sign of $k\delta + \Im \zeta$. This is the original RC32 interference identity, with its complete hidden term retained.

For the physical action $\mathcal{A} = kI/2 + iM$, put $z_\times = \langle h, \mathcal{A}y \rangle$, $w_\times = \langle y, \mathcal{A}h \rangle$. Orthogonality cancels the scalar term, giving

$$z_\times = iY\xi, \quad w_\times = iY\zeta, \quad \overline{z_\times} w_\times / Y^2 = \bar{\xi} \zeta. \quad (\text{SC3})$$

If $\xi = A + iB$, $\zeta = u + iv$, the real and imaginary parts are exactly $Au + Bv$ and $Av - Bu$. Thus the bilinear currents and centered current have a proved common scalar receiver while retaining their different signs.

Let I_K be the complete fixed kernel frame, $H_K = I_K^* G I_K$, $b_K = I_K^* G x$, $c_K = H_K^{-1} b_K$, $F_K = M I_K$, $A_K = I_K^* G F_K$, $s_K = x^* G F_K$. Then

$$h = I_K c_K, \quad Y = x^* G x - b_K^* H_K^{-1} b_K, \quad \zeta = (s_K c_K - c_K^* A_K c_K) / Y, \quad \chi = c_K^* A_K c_K / Y. \quad (\text{SC4})$$

The kernel projection formula proves the first two identities. Substitution of $y = x - I_K c_K$ proves the last two. In particular F_K retains its full codomain E ; it is not replaced by its kernel-to-kernel compression.

The original source decomposition gives

$$B_K := \|M|_K\|_{G \rightarrow G} \leq \|C\| + \epsilon \sqrt{\alpha}, \quad |\zeta| \leq B_K r_K, \quad |\chi| \leq B_K r_K^2. \quad (\text{SC5})$$

For $h \in K$, $|e^\dagger h| = |(P_K e)^\dagger h| \leq \sqrt{\alpha} \|h\|$. Apply the triangle inequality to $Mh = Ch + \epsilon f(e^\dagger h)$, then Cauchy-Schwarz to SC1. Thus $\log \max(1, B_K) = O(k \log(q+2))$ by RC25. The terminal visibility estimate TR3 bounds $r_K \leq \sqrt{\mathfrak{l}_{k,N}^{-1} - 1}$ with $-\log \mathfrak{l} = O(k \log(q+2))$. These are proved native bounds, with the original period guard.

2. Coherent complete-metric error, including the projection derivative

For two positive complete metrics G_0, G_1 , fix the same M, Λ, x, ω , and let

$$S = G_0^{-1/2} G_1 G_0^{-1/2}, \quad d = \frac{1}{2} \log(\lambda_{\max}(S) / \lambda_{\min}(S)), \quad B = B_K(G_0), \quad r = r_K(G_0). \quad (\text{SC6})$$

These are finite-dimensional statements on the actual metric family. Along a differentiable positive-metric path put $E_t = G_t^{-1} \dot{G}_t$; it is selfadjoint in G_t . Differentiation of $P_K = I_K(I_K^* G I_K)^{-1} I_K^* G$ gives

$$\dot{P}_K = P_K E_t P_B, \quad \dot{h} = P_K E_t y, \quad \dot{y} = -P_K E_t y, \quad \dot{Y} = \langle y, E_t y \rangle. \quad (\text{SC7})$$

For the last equation, both vector-derivative terms vanish by $y \perp K$. In the derivative of $b = \langle y, Mh \rangle$, the derivative of the first vector cancels the hidden part of the metric derivative. Exactly

$$\dot{b} = \langle y, E_t P_B Mh \rangle + \langle y, M P_K E_t y \rangle, \quad |\dot{\zeta}| \leq B_t(1 + 2r_t) \|E_t\|_{G_t}. \quad (\text{SC8})$$

Indeed the first term has magnitude at most $B_t r_t Y_t \|E_t\|$, and the second at most $B_t Y_t \|E_t\|$. The derivative of $1/Y_t$ contributes at most another $B_t r_t \|E_t\|$. Both surviving actions of M have arguments in the fixed full kernel.

Let $a_0 = \frac{1}{2}(\log \lambda_{\max} S + \log \lambda_{\min} S)$ and use the exact path $G_t = G_0^{1/2} \exp(t(\log S - a_0 I)) G_0^{1/2}$, $0 \leq t \leq 1$. Here a_0 is only this paragraph's metric scalar, not the monic-profile endpoint. The endpoint is $e^{-a_0} G_1$. Multiplying a whole metric by a positive scalar leaves its projections, SC1 and normalized current unchanged and multiplies the unnormalized current by that positive scalar. This proves the exact connecting map used for this path. Its relative eigenvalues lie in $[e^{-td}, e^{td}]$, so

$$\|E_t\|_{G_t} = d, \quad B_t \leq e^{td} B, \quad r_t^2 \leq e^{2td}(1 + r^2) - 1. \quad (\text{SC9})$$

The last bound compares $\|x\|_{G_t}^2$ and the complete minimum $Y_t = \min_{h \in K} \|x - h\|_{G_t}^2$ using the same two relative form bounds.

Define the explicitly evaluated integral

$$\begin{aligned} \mathcal{F}(d, r) &= \int_1^{e^d} [1 + 2\sqrt{(1 + r^2)s^2 - 1}] ds, \\ \mathcal{F}(d, r) &= e^d - 1 + e^d \sqrt{(1 + r^2)e^{2d} - 1} - r \\ &\quad - \frac{\text{arcosh}(e^d \sqrt{1 + r^2}) - \text{arcosh} \sqrt{1 + r^2}}{\sqrt{1 + r^2}}. \end{aligned} \quad (\text{SC10})$$

Differentiating the second expression with respect to its upper endpoint gives the first integrand; the lower-endpoint value is zero. The formula extends to $r = 0$ by continuity. Integrating SC8 using SC9 and $s = e^{td}$ proves

$$|\zeta_{G_1} - \zeta_{G_0}| \leq B\mathcal{F}(d, r), \quad |j_{G_1} - j_{G_0}| \leq 2B\mathcal{F}(d, r). \quad (\text{SC11})$$

This keeps the full change in metric and projection together. It does not bound independent perturbations of eigensector coordinates by the same number.

The bilinear currents have a complete disk estimate as well. Put $r_* = [e^{2d}(1 + r^2) - 1]^{1/2}$, $\delta_\zeta = B\mathcal{F}(d, r)$, and

$$\delta_\xi = 2d(|\omega| + e^d B)r_*(1 + r_*). \quad (\text{SC12})$$

Differentiating $r_t^2 = \|h_t\|^2/Y_t$ gives $|(r_t^2)'| \leq 2r_t(1 + r_t)\|E_t\|$. The two vector derivatives and metric derivative of $\langle h, Mh \rangle/Y$ give $|\dot{\chi}| \leq 2B_t r_t(1 + r_t)\|E_t\|$: each vector derivative is at most $B_t r_t Y \|E_t\|$, and the numerator metric and denominator terms total at most $2B_t r_t^2 Y \|E_t\|$. Integrating proves $|\xi_1 - \xi_0| \leq \delta_\xi$. Expanding the product difference now gives

$$\begin{aligned} &|\bar{\xi}_1 \zeta_1 - \bar{\xi}_0 \zeta_0| \\ &\leq |\xi_0| \delta_\zeta + |\zeta_0| \delta_\xi + \delta_\xi \delta_\zeta. \end{aligned} \quad (\text{SC13})$$

Its real and imaginary parts have this same radius.

For $0 \leq d \leq 1/4$, $\mathcal{F}(d, r) \leq 5(1 + r)d$. An explicit bound suffices: in its equivalent integral over $t \in [0, 1]$, use $e^{td} \leq 4/3$, $\sqrt{e^{2td} - 1} < 1$, and $\sqrt{e^{2td} r^2 + e^{2td} - 1} \leq (4/3)r + 1$. The integral is at most $[4 + (32/9)r]d \leq 5(1 + r)d$. Thus

$$|j_1 - j_0| \leq 10B(1 + r)d. \quad (\text{SC14})$$

If $(1 - \eta)G_0 \preceq G_1 \preceq (1 + \eta)G_0$, $0 \leq \eta \leq 1/8$, then $d \leq \text{arctanh } \eta \leq 2\eta$. An evaluated current margin $\mu > 0$ therefore retains half its margin when $\eta \leq \min\{1/8, \mu/[40B(1 + r)]\}$. This is an error certificate for an evaluated center, not an assertion of a native margin. By SC5, the sufficient logarithmic precision is $O(k \log(q + 2)) + \log^+(1/\mu)$.

3. The exact adjacent-cutoff polynomial in the original parameter

Let u be the next actual source column, and retain $G(t) = (G^{-1} + tuu^*)^{-1}$, $t \geq 0$; the adjacent native cutoffs are $t = 0, 1$. In this section u denotes that source column, not the boundary column in ST5. Define

$$\begin{aligned} u_K &= P_K u, \quad u_B = P_B u, \quad a = \|u_K\|_G^2, \quad \beta = \|u_B\|_G^2, \quad \rho = \langle u_B, x \rangle_G, \\ s &= t/(1 + \beta t). \end{aligned} \quad (\text{SC15})$$

Direct multiplication of inverse matrices, or the complete original ACC4–8 proof, gives

$$G(t) = G - \frac{s}{1 + as} G u u^* G, \quad h_s = h - s \rho u_K, \quad y_s = y + s \rho u_K.$$

$$\begin{aligned}\langle y_s, v \rangle_{G(t)} &= \langle y, v \rangle_G - s\bar{\rho}\langle u_B, v \rangle_G \quad (v \in E), \\ Y_s &= Y - s|\rho|^2 > 0.\end{aligned}\tag{SC16}$$

For the projection formula, $x - y_s \in K$, and the displayed covector annihilates K ; uniqueness of the orthogonal minimum proves it. To verify the covector, expand $y_s^*G(t)$, use $y^*Gu = \bar{\rho}$, $u_K^*Gu = a$, and cancel its $1 + as$ factor. Finally $|\rho|^2 \leq \beta Y$ gives $Y_s \geq Y/(1 + \beta t) > 0$.

It follows that the full unnormalized marked current is exactly $Q(s) = q_0 + q_1s + q_2s^2$, where

$$\begin{aligned}q_0 &= -2\Im[\omega Y - \langle y, Mh \rangle_G], \\ q_1 &= -2\Im[\rho\langle y, Mu_K \rangle_G - \omega|\rho|^2 + \bar{\rho}\langle u_B, Mh \rangle_G], \\ q_2 &= 2|\rho|^2\Im\langle u_B, Mu_K \rangle_G.\end{aligned}\tag{SC17}$$

Insert $My_s = My + s\rho Mu_K$ in SC16, then substitute $My = \omega x - Mh$. This proves all three coefficients and shows every non-eigenvalue action has its argument in the original full kernel. In the literal source parameter,

$$\Phi_{G(t)}(\Lambda x) = \frac{q_0 + (2\beta q_0 + q_1)t + (\beta^2 q_0 + \beta q_1 + q_2)t^2}{(1 + \beta t)^2}.\tag{SC18}$$

The denominator is positive. On the native step, $s \in [0, s_1]$, $s_1 = 1/(1 + \beta)$. The minimum of Q is attained at an endpoint or, only when $q_2 > 0$ and $0 < -q_1/(2q_2) < s_1$, at its vertex with value $(4q_0q_2 - q_1^2)/(4q_2)$. This follows by differentiating the polynomial. Thus at most three real evaluations decide positivity on the complete step; apply the same rule to $-Q$ for negativity. Coefficient radii δ_i bound its entire interval error by $\delta_0 + s_1\delta_1 + s_1^2\delta_2$. Direct differentiation also proves

$$\left(\frac{Q(s)}{Y - s|\rho|^2}\right)'' = \frac{2[q_2Y^2 + q_1Y|\rho|^2 + q_0|\rho|^4]}{(Y - s|\rho|^2)^3}.\tag{SC19}$$

The curvature sign is constant on this complete source step, including the case of zero curvature.

4. All coefficients of the imaginary bilinear current

For this paragraph let A be any fixed full action, and in the programme take $A = \mathcal{A}$. Put $\eta_K = \langle u_K, h \rangle_G$. Define

$$\begin{aligned}W_0 &= \langle y, Ah \rangle, \quad W_1 = -\rho\langle y, Au_K \rangle - \bar{\rho}\langle u_B, Ah \rangle, \\ W_2 &= |\rho|^2\langle u_B, Au_K \rangle, \\ Z_0 &= \langle h, Ay \rangle, \\ Z_1 &= \rho\langle h, Au_K \rangle + aZ_0 - \bar{\rho}\langle u_K, Ay \rangle - \bar{\eta}_K\langle u, Ay \rangle, \\ Z_2 &= a\rho\langle h, Au_K \rangle - |\rho|^2\langle u_K, Au_K \rangle \\ &\quad - \rho\bar{\eta}_K\langle u, Au_K \rangle + a\bar{\rho}\langle u_B, Ay \rangle, \quad Z_3 = aW_2.\end{aligned}\tag{SC20}$$

All inner products here are the initial G -products. Expanding SC16 proves $w_\times(s) = \sum_{v=0}^2 W_v s^v$. For the reverse product, direct expansion gives the covector

$$h_s^*G(t) = \frac{h^*G + s(ah^*G - \bar{\rho}u_K^*G - \bar{\eta}_K u^*G) + s^2 a\bar{\rho}u_B^*G}{1 + as}.$$

Multiplying by $A(y + s\rho u_K)$ proves $z_\times(s) = \sum_{u=0}^3 Z_u s^u/(1 + as)$ and every coefficient SC20. The highest coefficient of the product numerator is $Z_3 W_2 = a|W_2|^2$, which is real. Therefore, with

$$I_j = \Im \sum_{u+v=j, 0 \leq u \leq 3, 0 \leq v \leq 2} \bar{Z}_u W_v \quad (0 \leq j \leq 4),$$

$$\Im(\overline{z_\times(t)} w_\times(t)) = \frac{\sum_{j=0}^4 I_j t^j (1 + \beta t)^{4-j}}{[1 + (a + \beta)t](1 + \beta t)^3}.$$

(SC21)

Both denominator factors are positive. A nonzero imaginary numerator has at most four real zeros counted with multiplicity. Its sign on a finite step is determined by this actual real quartic; an identically zero numerator is retained. The real numerator retains the fifth-degree term unless that explicit coefficient vanishes.

5. Exact finite examples and sharpness

The following examples test the scope of the algebraic statements. They are not native-period numerical samples. First take $G = I_3$,

$$C = \begin{pmatrix} 0 & -3/5 & 0 \\ -3/5 & 0 & -4/5 \\ 0 & -4/5 & 0 \end{pmatrix}, \quad M = C + \frac{10}{3}e_2e_1^*,$$

$$x = (3/10, i/2, 2/5)^T, \quad v = (0, 4i/5, 3/5)^T, \quad K = \mathbb{C}v.$$

Multiplication gives $Mx = -ix$, $C = C^*$, $\|C\| = 1$, $\|x\|^2 = 1/2$, $\|h\|^2 = 256/625$, $Y = 113/1250$, $\langle h, Mh \rangle = 0$, and $\langle y, Mh \rangle = -64i/625$. Thus the full eigenclass current is 1, whereas $\Phi_G = -3/125$. For the unchanged M, K, x , use the source column $u = v + y$. Substitution in SC17-18 gives

$$\Phi_{G(t)} = \frac{10(4181t - 3750)}{(113t + 1250)^2}, \quad t_* = 3750/4181 \in (0, 1),$$

$$\Phi_{G(1)} = 4310/1857769 > 0. \quad (\text{SC22})$$

This computes a crossing with a fixed eigenclass, zero hidden self-current and unchanged first outgoing direction. It does not infer a native sign from those properties.

For two crossings take $M = \begin{pmatrix} 0 & 0 & 0 \\ i/2 & -3i/256 & 0 \\ -67i/256 & 0 & -3i/256 \end{pmatrix}$, $x = (0, 1, 1)^T$, $K = \mathbb{C}e_1$, $u = (1, 1, 0)^T$. Then $Mx = -3ix/256$, and SC18 gives $\Phi_{G(t)} = (7t - 1)(5t - 3)/[64(1 + t)^2]$, with zeros $1/7, 3/5$. The degree-two bound is attained. For the same x, K, u , put $A = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}$. SC20 gives $z_\times = t^3/[(1 + t)^2(1 + 2t)]$, $w_\times = -t/(1 + t)^2 - it/(1 + t)$, and

$$\Im(\bar{z}_\times w_\times) = -\frac{t^4}{(1 + 2t)(1 + t)^3}. \quad (\text{SC23})$$

This attains degree four. All these substitutions are verified as exact rational identities by the accompanying script, alongside the general projection, coefficient and differential identities.

6. New return through the mixed probes and all four original cutoffs

At each original cutoff, MP19-24 and SC2 give the exact additional receiver

$$\boxed{\begin{aligned} \frac{\widehat{b}^* \mathcal{C}_{\text{ord}} \widehat{b}}{s_{\text{mix}}} &= -2\Im\omega + 2\Im\zeta_G, \\ \widehat{b} &= Q_B^{1/2} \Lambda x / \sqrt{Y}. \end{aligned}} \quad (\text{SC24})$$

Here $s_{\text{mix}} = (\dim B - \text{tr } A)/(q - 2)$ is MP5; it is different from the source-activation parameter in SC15. Both are explicitly specified. Therefore the complete ratio on the left, recalculated from the same coherently varied metric, obeys SC11 and SC14. This carries the new stability estimate through the mixed construction without paying the norm of the large current or assigning independent perturbations to its terms. The original MP inverse still bounds errors in an independently supplied Hermitian mixed-product operator by $1/s_{\text{mix}} < 6$. These two statements address their stated error models.

For each of the two original adjacent-cutoff paths, calculate the three actual coefficients SC17. Denote them q_i^- at $q - 1 \rightarrow q$, q_i^+ at $2q - 1 \rightarrow 2q$, and denote their actual source values by β_- , β_+ . The full unnormalized four-cutoff current is exactly

$$\boxed{\begin{aligned} \mathcal{R}\Phi &= 2q_0^- + \frac{q_1^-}{1 + \beta_-} + \frac{q_2^-}{(1 + \beta_-)^2} \\ &\quad - 2q_0^+ - \frac{q_1^+}{1 + \beta_+} - \frac{q_2^+}{(1 + \beta_+)^2}. \end{aligned}} \quad (\text{SC25})$$

This is substitution of the literal endpoints $t = 0, 1$ in SC18 with signs $+, +, -, -$. It retains the unnormalized class and the original metric at each endpoint. The two real quartics SC21 likewise give both endpoint imaginary bilinear currents with their complete denominators. No sign of these original coefficients has been selected from magnitude bounds. The unfinished calculation is now their signed native value, with exact finite equations and coherent full-metric precision requirements already supplied.

Provenance: the stable-sign continuation supplied on 22 September is rederived in full above, including the three exact examples. The source-step data and complete complex currents are the original ACC1-16; the rectangular hidden-action and period bounds are the pinned RC22-44 and TR1-24 proofs linked above. The new receiver SC24 and four-cutoff formula SC25 apply them to MP19-24. Human-source provenance for the inherited arithmetic source remains in those complete published proofs. No new human literature or native-period numerical evaluation is claimed.

Original arithmetic evolution on a subexponential clock

22 September 2026. Independent derivation EE1–48. The incoming continuation is checked here on the unchanged original simple-quartet family, fixed arithmetic source, period and branch. Every estimate below uses the complete original source minimum and observation. The central additions are a growing source-jet estimate, characteristic-polynomial control of repeated rank-one feedback, and an actual inverse formula for the measured propagator. These give the previously inaccessible lower time plateau. They do not determine the residual terminal current or establish RH.

1. Exact original inputs, conventions and source locators

Let $k \geq 17$, $k \equiv 1 \pmod{4}$, $q = (k+1)^2$, $q_0 = (k-7)^2$, and $\Delta = q - q_0 = 16k - 48$. Retain

$$Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad 0 < \delta < 1/2, \quad \gamma > 2, \quad (EE1)$$

$$E = \mathbb{C}[y]/Q_k.$$

The four cutoffs are $N = q - 1, q, 2q - 1, 2q$, with return signs $+, +, -, -$. Write $s_N = (N + 1 - q)/q$. All estimates also hold throughout this original degree window when the stated finite guards hold.

Let G_N be the full source-minimum metric, $\Lambda : E \rightarrow B$ the unchanged observation and

$$Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L_N = G_N^{-1} \Lambda^* Q_{B,N}, \quad J = L_N Q_{B,N}^{-1/2}, \quad (EE2)$$

$$P = J J^\dagger = L_N \Lambda, \quad \Pi = I - P.$$

The dagger is the original G_N -adjoint. Thus $J^\dagger J = I_B$. All displayed compressed matrices act in these isometric coordinates; transporting them by $Q_{B,N}^{1/2}$ gives precisely the original observed-metric operators.

The original orthogonal polynomials p_n , their squared norms ω_n , and the complete relation minima give exactly

$$M = C + R, \quad C = C^\dagger, \quad R = \epsilon f e^\dagger, \quad e^\dagger f = 0, \quad \|e\| = \|f\| = 1,$$

$$e = [p_N]/\sqrt{E_N}, \quad f = [p_{N+1}]/\sqrt{F_N}, \quad \epsilon = \sqrt{E_N F_N}/\omega_N. \quad (EE3)$$

These are OCP5–7, rederived there by exact projection of real multiplication onto the complete minimum-lift subspace. The isometry and measured/intrinsic distinction are CE2–3, CE24–29.

Put $u = J^\dagger e$, $v = J^\dagger f$, $a = u^* u$, $d = v^* v$, $r = u^* v$, $R_B = \epsilon v u^*$, and $\epsilon_B = \epsilon \sqrt{ad}$. The complete two-column certificate is

$$0 \prec \Theta I_2 \preceq \begin{pmatrix} a & r \\ \bar{r} & d \end{pmatrix} \preceq I_2, \quad (EE4)$$

$$ad - |r|^2 \geq \Theta.$$

The determinant assertion is retained explicitly: the matrix lower bound alone would only give determinant at least Θ^2 . It follows that

$$\epsilon \sqrt{\Theta} \leq \epsilon_B \leq \epsilon, \quad |r| = |e^\dagger \Pi f| \leq \sqrt{\alpha}, \quad (EE5)$$

$$\alpha = \|\Pi e\|^2.$$

Here $e^\dagger f = 0$ was used, with its sign: $r = -e^\dagger \Pi f$.

The proved source cost and monic inputs are RC22–25:

$$\ell_k \|p\|_\sigma^2 \leq \|p\|_{\mu_k}^2 \leq u_k \|p\|_\sigma^2 \quad (\deg p \leq 2q + 2),$$

$$\log(u_k/\ell_k) = O_h(k + \log q),$$

$$\log \epsilon = q\psi(s_N) + O_{h,\varpi}(k \log(q+2)), \quad -\log \Theta = O_{h,\varpi}(k \log(q+2)), \quad (EE6)$$

$$\alpha \leq e^{-2q\psi(s_N) + O_{h,\varpi}(k \log(q+2))}.$$

The Gamma measure is

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad h_n = \sqrt{2\pi} n!(1/2)_n. \quad (EE7)$$

Choose throughout

$$c = \max\{1, 2(N+1)\sqrt{u_k/\ell_k}\}. \quad (\text{EE8})$$

It bounds the full arithmetic degree- N Jacobi compression Y_N , its compression C , and $C_B = J^\dagger C J$. Indeed the Gamma orthonormal recurrence has off-diagonal coefficient $\sqrt{(n+1)(n+1/2)}$; the two neighboring diagonals bound real multiplication on degree at most N by $2(N+1)$. Applying the same source comparison in degrees $N, N+1$ proves EE8. This stronger choice is necessary when telescoping through Y_N .

The human recurrence and generating function are T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, DLMF18.22.8 and 18.23.7, with $\lambda = 1/4$, $\phi = \pi/2$, $x = y/2$. Their original equation TeX was read and retained in the preceding CE source directory. This reading covers those formulas, not the entire chapter. All programme-specific transfers are proved below.

2. Complete growing-jet estimates

The conductor from OCP11-14 is

$$\mathcal{T}p = \sum_{a,b=0}^8 a_{ab}p(y + \sigma_{ab}), \quad \sigma_{ab} = (2b-8)\gamma - i(2a-8)\delta.$$

Its first nonzero moment order $v_0 \leq 80$ is fixed, and its nonzero leading coefficient on degree n is

$$\tau_n = i^{-v_0} \mu_{v_0} \binom{n}{v_0}.$$

The exact lower-quotient map $C_A[p] = [\mathcal{T}p]_{Q_{k-8}}$ obeys $K = \ker \Lambda \subset \ker C_A$. Thus every minimum-lift polynomial of an original kernel class satisfies

$$\mathcal{T}p \in Q_{k-8}\mathcal{P}_{D-q_0}, \quad D = N - v_0. \quad (\text{EE9})$$

No equality of the two kernels is used. The full finite Gamma conductor norm is

$$\|\mathcal{T}\|_{\sigma, \mathcal{P}_N \rightarrow \mathcal{P}_D} \leq \mathcal{F}_N = A_1 e^{1+2\pi\gamma} (N+1)(2N+1)^{4\delta+1/2}, \quad A_1 = \sum |a_{ab}|. \quad (\text{EE10})$$

It follows by multiplying the original generating function by each $e^{\sigma_{ab} \arctan z}$, applying Cauchy's coefficient bound at $N/(N+1)$, and summing all diagonals; the full derivation is OCP13.

For an integer $0 \leq J \leq k$, retain the guard $J \leq D - q_0$. It holds at all four cutoffs: $D - q_0 - J \geq 15k - 49 - v_0 \geq 0$ for the stated $k \geq 17, v_0 \leq 80$. Let $\nu_j^{-, \sigma}$ and $\nu_j^{+, \sigma}$ be the complete monic minima for $|Q_{k-8}|^2 d\sigma$ and $|Q_k|^2 d\sigma$. Their original roots are all retained.

Here is a finite coefficient lemma used twice. If two monic polynomial families of degree at most D have all roots of modulus at most $B \geq 1$, the coefficient l places below their leading terms is at most $(DB)^l$. Conversion between the two families therefore costs at most $(2DB)^l$: recursively cancel the leading terms; with $A = DB$, the coefficient bound satisfies

$$b_l \leq A^l + \sum_{j=0}^{l-1} b_j A^{l-j}, \quad b_0 = 1,$$

Direct triangular subtraction and induction therefore give $b_l \leq A^l + \sum_{j=0}^{l-1} (2A)^j A^{l-j} \leq (2A)^l$. This last sum is $A^l(1 + 2^l - 1)$. The bound includes all lower coefficients and needs no location on the real line for the roots of Q .

The residual orthogonal polynomials for $|Q|^2 d\sigma$ do have real roots of modulus at most $2(D+1)$: their zeros are eigenvalues of the compressed multiplication matrix on $Q\mathcal{P}_{j-1}$, which is a compression of Gamma multiplication on degree at most $D-1$. The ordinary Gamma polynomials have the same bound. The preceding coefficient lemma therefore applies with

$$Z_- = 2(D+1) \max\{1, R_{k-8}, 2(D+1)\},$$

$$(\mathcal{E}_{N,J}^-)^2 = \sum_{i=0}^J \sum_{j=i}^J Z_-^{2(j-i)} \frac{h_{D-j}}{\nu_{D-q_0-i}^{-, \sigma}}. \quad (\text{EE11})$$

Expand every orthonormal relation row $Q_{k-8}U_{D-q_0-i}/\sqrt{\nu_{D-q_0-i}^{-(b),\sigma}}$ in the original Gamma basis. Its coefficient at degree $D-j$ is zero for $j < i$, and at most the square root of the corresponding summand in EE11 otherwise. All other relation rows have degree below $D-J$, so contribute exactly zero. The Frobenius bound on this complete finite matrix proves that $\mathcal{E}_{N,J}^-$ bounds the full leading-jet map on the lower ideal.

Let A_J be the conductor matrix between the normalized Gamma top jets of degrees $N, \dots, N-J$ and $D, \dots, D-J$. Its diagonal entries are $\tau_{N-j}\sqrt{h_{D-j}/h_{N-j}}$, its norm is at most $\mathcal{F}_N$, and it is triangular because the conductor lowers every degree by v_0 . Put

$$\begin{aligned} \beta_{N,J} &= \\ \max \left\{ 1, \mathcal{F}_N \max_{0 \leq j \leq J} \frac{\sqrt{h_{N-j}}}{|\tau_{N-j}|\sqrt{h_{D-j}}} \right\}, \\ \mathcal{T}_{N,J} &= \beta_{N,J} \sum_{\ell=0}^J [(J+1)\beta_{N,J}]^\ell. \end{aligned} \quad (\text{EE12})$$

To prove this inverse bound, divide A_J by $\mathcal{F}_N$, split its diagonal and strictly triangular part, and invert $I + H$ by $\sum_{\ell=0}^J (-H)^\ell$. Each entry of $A_J/\mathcal{F}_N$ has modulus at most one, its inverse diagonal norm is at most $\beta_{N,J}$, and $\|H\| \leq (J+1)\beta_{N,J}$. Hence $\|(A_J/\mathcal{F}_N)^{-1}\| \leq \mathcal{T}_{N,J}$. For every p satisfying EE9,

$$\|\text{top}_{J+1}^\sigma p\| \leq \mathcal{T}_{N,J} \mathcal{E}_{N,J}^- \|p\|_\sigma. \quad (\text{EE13})$$

The apparent missing conductor factor cancels exactly: applying the inverse gives $1/\mathcal{F}_N$, whereas $\|\mathcal{T}p\|_\sigma \leq \mathcal{F}_N \|p\|_\sigma$.

Transfer the entire affine leading-jet fibre once. For every polynomial p ,

$$\begin{aligned} \ell_k \text{dist}_\sigma(p, \mathcal{P}_{N-J-1})^2 \\ \leq \text{dist}_{\mu_k}(p, \mathcal{P}_{N-J-1})^2 \\ \leq u_k \text{dist}_\sigma(p, \mathcal{P}_{N-J-1})^2. \end{aligned} \quad (\text{EE14})$$

Taking infima of the same source inequalities proves both sides. Let $\Phi_{N,J}$ be the orthonormal arithmetic polynomial frame $(p_N/\sqrt{\omega_N}, \dots, p_{N-J}/\sqrt{\omega_{N-J}})$. Let P_K^{pol} project onto the complete minimum-lift kernel in $\mathcal{P}_N$, and P_I onto the full relation ideal $Q_k \mathcal{P}_{N-q}$. Applying EE14 to EE13 on the former subspace, then taking the adjoint of its top-jet map, proves

$$\begin{aligned} \|P_K^{\text{pol}} \Phi_{N,J}\| &\leq L_{K,N,J} := \sqrt{u_k/\ell_k} \\ &\mathcal{T}_{N,J} \mathcal{E}_{N,J}^-. \end{aligned} \quad (\text{EE15})$$

The metric ratio occurs once for this whole map, not at every jet order.

For the original upper relation ideal the same coefficient proof gives

$$Z_+ = 2(N+1) \max\{1, R_k, 2(N+1)\},$$

$$\begin{aligned} (\mathcal{E}_{N,J}^+)^2 &= \sum_{i=0}^{\min(J, N-q)} \sum_{j=i}^J \\ &Z_+^{2(j-i)} \frac{h_{N-j}}{\nu_{N-q-i}^{+(b),\sigma}}, \\ L_{I,N,J} &:= \sqrt{u_k/\ell_k} \mathcal{E}_{N,J}^+, \\ \|P_I \Phi_{N,J}\| &\leq L_{I,N,J}. \end{aligned} \quad (\text{EE16})$$

For $N = q-1$ the ideal and this sum are zero. Every absent row has degree below the jet and contributes zero; no relation is omitted on the other three cutoffs.

The authoritative monic input is LRS11, with its exact Gamma cancellation and original lower roots; its receiver RC23 states

$$\begin{aligned} \log \frac{\nu_j^{(b),\sigma}}{h_{Q_b+j}} &= 2Q_b \psi(j/Q_b) + O_h(\log q), \quad 0 \leq j \leq 2Q_b, \\ b &= k, k-8. \end{aligned} \quad (\text{EE17})$$

All indices in EE11 and EE16 lie in that interval eventually, including zero. Replacing j by $j - i$ shifts the argument by at most J/Q_b . The exact endpoint derivative $\psi'(t) = \frac{1}{4} \log t + \log(4/\pi) + O(\sqrt{t})$, and bounded derivatives away from zero, give total exponent displacement $O((k+J+1) \log(q+2))$, including $q - q_0 = O(k)$. Also

$$\frac{h_{N-j}}{h_{N-i}} \leq 1 \quad (j \geq i), \quad \log Z_{\pm} = O_h(\log q).$$

The normalized conductor diagonal satisfies exactly

$$|\tau_n|^2 \frac{h_{n-v_0}}{h_n} = \frac{|\mu_{v_0}|^2}{(v_0!)^2} \prod_{l=0}^{v_0-1} \frac{n-l}{n-l-1/2},$$

so its positive and negative logarithms are bounded on the fixed data. Thus $\log \beta_{N,J} = O_{h,\varpi}(\log q)$ and $\log \mathcal{T}_{N,J} = O_{h,\varpi}((J+1) \log(q+2))$. Counting the at most $(J+1)^2$ summands now proves the uniform estimate

$$L_{K,N,J} + L_{I,N,J} \leq \exp[-q\psi(s_N) + O_{h,\varpi}((k+J+1) \log(q+2))]. \quad (\text{EE18})$$

This is a proved growing-jet bound derived from the complete source, rather than a fixed-order estimate extrapolated to $J = k$.

In polynomial coordinates the observed orthogonal projection is $I - P_I - P_K^{\text{pol}}$. The latter two projections have orthogonal ranges. Therefore

$$[1 - L_{K,N,J}^2 - L_{I,N,J}^2]I \preceq \Phi_{N,J}^*(I - P_I - P_K^{\text{pol}})\Phi_{N,J} \preceq I. \quad (\text{EE19})$$

This is the Gram of the specified source columns after their actual observation, not a replacement kernel frame.

Put $S = I - P_I$, $z = p_N/\sqrt{\omega_N}$, and $\vartheta = \|Sz\|^2 = E_N/\omega_N$. The isometry from source quotient to $\mathcal{SP}_N$ sends e to $Sz/\sqrt{\vartheta}$ and C to $SY_N S$ restricted to that subspace. Since Y_N is tridiagonal, $Y_N^j z$ belongs to the final $j+1$ polynomial directions. Telescoping gives, for $j \leq J$,

$$\|(SY_N S)^j Sz - Y_N^j z\| \leq (j+1)L_{I,N,J}c^j.$$

The initial error is L_I ; each subsequent step adds at most $L_I c^j$, since $\|Y_N\| \leq c$. Taking P_K^{pol} and using EE15 proves

$$\|\Pi C^j e\| \leq c^j \Xi_{N,J}, \quad \Xi_{N,J} := \frac{L_{K,N,J} + (J+1)L_{I,N,J}}{\sqrt{\vartheta}}. \quad (\text{EE20})$$

The exact monic formula OCP5, or EE16 at $J = 0$, gives $\vartheta \geq 1/2$ eventually. Since $\min_{0 \leq s \leq 1+1/q} \psi(s) > 0$ eventually, EE6 and EE18 yield

$$\epsilon \Xi_{N,k} \leq e^{O_{h,\varpi}(k \log(q+2))}. \quad (\text{EE21})$$

3. Full characteristic feedback and a finite all-time remainder

Write $R_{\text{root}} = k\sqrt{\delta^2 + \gamma^2}$ and $\rho = 8q \max(1, c, R_{\text{root}})$. The determinant identity, proved by multiplying $z - C - \epsilon f e^\dagger$ by $(z - C)^{-1}$, is

$$1 - \epsilon e^\dagger (z - C)^{-1} f = \frac{Q_k(z)}{\det(z - C)}. \quad (\text{EE22})$$

On $|z| = \rho$, use the logarithms of $1 - \lambda/z$, continuous from infinity, for all q original roots and all q eigenvalues of C . Since every ratio has modulus at most $1/(8q)$,

$$\left| \log \frac{Q_k(z)}{\det(z - C)} \right| \leq 2q \frac{1/(8q)}{1 - 1/(8q)} \leq 2/7.$$

The resolvent identity then gives

$$\begin{aligned} \|(z - M)^{-1} - (z - C)^{-1}\| &\leq \frac{\epsilon e^{2/7}}{\rho^2(1 - 1/(8q))^2} \\ &< \frac{2\epsilon}{\rho^2}. \end{aligned} \quad (\text{EE23})$$

No diagonalization of M occurs; its full characteristic polynomial includes all multiplicities.

On the same circle, $|1 - Q_k / \det(z - C)| \leq e^{2/7} - 1 < 1$. Its Laurent coefficients give

$$g_j = \epsilon e^\dagger C^j f, \quad g_0 = 0, \quad |g_j| \leq \rho^{j+1}. \quad (\text{EE24})$$

Cauchy's integral formula applied to EE23 gives $\|M^n - C^n\| \leq 2\epsilon\rho^{n-1}$. For real t , e^{itC} is unitary and $\|e^{itC} - I\| \leq c|t|$. Remove the exact $n = 1$ term itR before summing:

$$\|e^{itM} - I - itR\| \leq F(|t|) := c|t| + \frac{2\epsilon}{\rho}(e^{\rho|t|} - 1 - \rho|t|). \quad (\text{EE25})$$

This finite inequality holds at every real time. Its small relative error is used only on the explicit clock below.

4. Actual compressed spectrum and intrinsic evolution

Put $D_B = J^\dagger M J = C_B + \epsilon v u^*$. For $0 \leq j \leq k$, the same telescoping as EE20, now in E , gives

$$\|(PCP)^j P e - C^j e\| \leq (j+1)c^j \Xi_{N,k}.$$

Taking its inner product with f and using selfadjointness of C_B proves

$$|\epsilon u^* C_B^j v - g_j| \leq \epsilon(j+1)c^j \Xi_{N,k}. \quad (\text{EE26})$$

Thus the growing-jet estimate is transported through the original observation, rather than assigning the full characteristic polynomial to D_B .

Let

$$H_* = \rho + c + (k+1)\epsilon \Xi_{N,k}, \quad r_B = \max\{2c, 4(k+2)H_*, (8\epsilon_B c^{k+1})^{1/(k+2)}\}. \quad (\text{EE27})$$

The first $k+1$ terms of $\epsilon u^*(z - C_B)^{-1}v$ have modulus sum at most $\sum_{j=0}^k (H_*/|z|)^{j+1} < 1/4$ when $|z| \geq r_B$. For $j \geq 1$, EE24 and EE26 are bounded by H_*^{j+1} ; the same bound for $j = 0$ follows from $g_0 = 0$. The complete tail is at most

$$\frac{\epsilon_B c^{k+1}}{|z|^{k+2}(1 - c/|z|)} \leq \frac{2\epsilon_B c^{k+1}}{|z|^{k+2}} \leq 1/4.$$

The determinant lemma for D_B therefore excludes every eigenvalue with $|z| \geq r_B$. Moreover

$$\log(1 + r_B) = O_{h,\omega}(k \log(q+2)). \quad (\text{EE28})$$

For its third term, $\log \epsilon_B / (k+2) = O(k)$ by EE6; all other terms use EE8 and EE21.

Take $n_B = \dim B$ and $\rho_B = 8n_B \max(1, c, r_B)$. Repeating the complete proof of EE23-25 for the actual characteristic polynomial of D_B gives

$$\|e^{itD_B} - I - itR_B\| \leq F_B(|t|) := c|t| + \frac{2\epsilon_B}{\rho_B} (e^{\rho_B|t|} - 1 - \rho_B|t|). \quad (\text{EE29})$$

Here $R_B^2 = (\epsilon r)R_B$; the proof does not assume that the observed rank-one map is square-zero.

5. Hidden evolution and the measured inverse

For $0 \leq J \leq k$, expand $e^\dagger e^{itC} \Pi$ through order J , use EE20 on the initial terms, and use $\|C\| \leq c$ on the tail. For $|t| \leq T$,

$$\|e^\dagger e^{itC} \Pi\| \leq E_K(T, J) := e^{cT} \left[\Xi_{N,J} + \frac{(cT)^{J+1}}{(J+1)!} \right]. \quad (\text{EE30})$$

The full scalar Duhamel equation for $F_K(t) = e^\dagger e^{itM} \Pi$ has free term $e^\dagger e^{itC} \Pi$ and convolution kernel $i\epsilon e^\dagger e^{itC} f$, with the orientation changed for negative t . By EE24 its absolute integral over length T is at most

$$\kappa_F(T) = e^{\rho T} - 1 - \rho T.$$

Taking the supremum in the exact equation, whenever $\kappa_F(T) < 1$, gives $\sup \|F_K(t)\| \leq E_K / (1 - \kappa_F)$. Substitute this in the vector Duhamel equation, using unitarity of e^{itC} , to obtain

$$\sup_{|t| \leq T} \|(e^{itM} - I)\Pi\| \leq d_K(T, J) := cT + \frac{\epsilon T E_K(T, J)}{1 - \kappa_F(T)}. \quad (\text{EE31})$$

This is a bound on the complete hidden source, with every feedback return included.

Set

$$\mathcal{L}_k = k \log(q+2) \log \log(q+e^e), \quad T_k = e^{-\mathcal{L}_k}. \quad (\text{EE32})$$

Because $\log c, \log \rho = O(k + \log q)$, $\log \rho_B = O(k \log q)$ and EE21 holds, the finite guards $\rho_B T_k < 1$, $\kappa_F(T_k) < 1/2$, $d_K(T_k, k) < 1/2$ all hold eventually. Direct substitution proves

$$d_K(T_k, k) \leq e^{-\mathcal{L}_k + O_{h,\infty}(k \log(q+2))}. \quad (\text{EE33})$$

For the Taylor tail, its logarithm is at most

$$q\psi(0) - (k+2)\mathcal{L}_k + O_h(k^2 + k \log q) - \log((k+1)!),$$

which is smaller than $-\mathcal{L}_k - Ck \log q$ for every fixed C eventually. This calculation is why the growing order is required.

Let $U(t) = J^\dagger e^{itM} J$. Decompose e^{-itM} on the exact orthogonal sum $PE \oplus \Pi E$:

$$e^{-itM} = \begin{pmatrix} A_- & B_- \\ C_- & D_- \end{pmatrix}.$$

EE31 gives $\|D_- - I\|, \|B_-\| \leq d_K(T, k)$, and EE25 gives $\|C_-\| \leq T\epsilon + F(T)$. If $d_K < 1$, block elimination gives a factorization with invertible lower block D_- . Since the whole exponential is invertible, its Schur complement is invertible; the upper block of its inverse is $U(t)$. Therefore

$$\boxed{U(t)^{-1} = U(-t) - B_- D_-^{-1} C_-},$$

$$\boxed{\|U(t)^{-1} - U(-t)\| \leq \frac{d_K(T, k)}{1 - d_K(T, k)} [T\epsilon + F(T)], \quad |t| \leq T.} \quad (\text{EE34})$$

This is the actual measured inverse. The second term generally does not vanish.

For a uniform relative norm estimate on $0 < |t| \leq T_k$, use EE34 with $T = |t|$, not with T_k in its numerator. All finite bounds are nondecreasing in T while their denominators remain positive. This keeps its factor $|t|\epsilon$ before comparison with the shear; replacing it prematurely by $T_k\epsilon$ would not prove uniformity near zero.

Jacobi's complementary determinant identity follows here from the same block factorization:

$$\det U(t) = \det(e^{itM}) \det D_- = \det D_-,$$

since $\text{tr } M$ is the sum of all original roots and is zero. The logarithm of $D_- = I + E$ has its convergent power series when $\|E\| < 1$. Thus, without choosing an unproved kernel-rank identification,

$$|\log \det U(t)| \leq (\dim K)[- \log(1 - d_K(T, k))] \leq q[- \log(1 - d_K(T, k))]. \quad (\text{EE35})$$

The logarithm is the branch continuous from zero. It tends to zero uniformly on the clock. If the programme's stated $\dim K = m$ is used, its sharper prefactor is exactly m .

6. Uniform shear law and both original endpoint pairs

Put $z_B = \epsilon r$, $h_B = |z_B|/\epsilon_B$. By EE5-6,

$$h_B \leq \sqrt{\alpha/\Theta} \leq e^{-q\psi(s_N) + O_{h,\infty}(k \log(q+2))}.$$

A rank-one map of norm one and trace ζ has matrix $\begin{pmatrix} \zeta & \sqrt{1-|\zeta|^2} \\ 0 & 0 \end{pmatrix}$ in a suitable orthonormal basis: take its image unit vector first and remove the trace component from its input unit vector. Its distance to $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is at most $2|\zeta|$, because $1 - \sqrt{1-|\zeta|^2} \leq |\zeta|$. The latter shear has norm

$$S(x) = \frac{\sqrt{x^2 + 4} + x}{2}, \quad S(x) \geq 1, \quad S(x) \geq x. \quad (\text{EE36})$$

Its two singular values follow by solving the quadratic characteristic polynomial of its positive Gram, whose determinant is one.

For $T = |t| \leq T_k$, EE25 divided by $S(T\epsilon_B)$ costs at most

$$cT + \frac{\epsilon}{\epsilon_B} \rho T e^{\rho T}.$$

The shear replacement costs at most $2h_B$. EE29 costs at most $cT + \rho_B T e^{\rho_B T} + 2h_B$. The measured inverse correction EE34, with this same T , costs at most

$$\frac{d_K(T, k)}{1 - d_K(T, k)} \left[\frac{\epsilon}{\epsilon_B} + cT + \frac{\epsilon}{\epsilon_B} \rho_B T e^{\rho_B T} \right].$$

Here $\epsilon/\epsilon_B \leq \Theta^{-1/2}$ is retained. Each expression is at most $\eta_k = \exp[-\mathcal{L}_k + O_{h,\varpi}(k \log(q+2))]$; the h_B term is even smaller since $\mathcal{L}_k = o(q)$ and $\psi(1) > 0$. Consequently, simultaneously for both time signs and all four cutoffs,

$$\begin{aligned} \|U(t)\| &= S(|t|\epsilon_B)(1 + O(\eta_k)), \\ \|U(t)^{-1}\| &= S(|t|\epsilon_B)(1 + O(\eta_k)), \\ \|e^{itD_B}\| &= S(|t|\epsilon_B)(1 + O(\eta_k)), \\ \|e^{-itD_B}\| &= S(|t|\epsilon_B)(1 + O(\eta_k)). \end{aligned} \tag{EE37}$$

This proves measured invertibility on the entire interval and its least singular value, in the original metric.

For $x > 0$, $0 \leq \log S(x) - (\log x)_+ \leq \log 2$. Together with EE6 and EE5, EE37 proves

$$\log s_{\max} U(t) = (\log |t| + q\psi(s_N))_+ + O(k \log q),$$

$$\log s_{\min} U(t) = -(\log |t| + q\psi(s_N))_+ + O(k \log q). \tag{EE38}$$

Both statements also hold for e^{itD_B} . Put $a_0 = \psi(0)$, $a_1 = \psi(1)$. The endpoint expansions in RC8-10 give $q\psi(1/q) = qa_0 + O(\log q)$ and $q\psi(1+1/q) = qa_1 + O(1)$. The positive-part map is 1-Lipschitz. Thus for every $b > 0$, with $e^{-bq} \leq T_k$ eventually,

$$\mathcal{R} \log s_{\max} U(e^{-bq}) = 2q[(a_0 - b)_+ - (a_1 - b)_+] + O_{h,\varpi}(k \log q),$$

$$\mathcal{R} \log s_{\min} U(e^{-bq}) = -2q[(a_0 - b)_+ - (a_1 - b)_+] + O_{h,\varpi}(k \log q),$$

$$\mathcal{R} \log \text{cond } U(e^{-bq}) = 4q[(a_0 - b)_+ - (a_1 - b)_+] + O_{h,\varpi}(k \log q). \tag{EE39}$$

All four terms are present. The intrinsic operator has the same formulas. In particular the lower plateau $0 < b < a_1$, and also the moving time T_k , give $4(a_0 - a_1)q + O(k \log q)$. The central scalar $e^{kt/2}$ of the physical action changes no condition number; for four-cutoff norms it cancels because the four signs sum to zero.

7. Growing exterior ranks and the actual terminal class

Let $r_k = \lceil 2q/\mathcal{L}_k \rceil = o(k)$. The exact identity

$$M^n - C^n = \sum_{j=0}^{n-1} M^{n-1-j} R C^j$$

shows that $\sum_{n=1}^{r_k} (it)^n (M^n - C^n)/n!$ has range contained in $\text{span}\{f, Mf, \dots, M^{r_k-1}f\}$, and hence rank at most r_k . EE23 bounds its remaining exponential tail by

$$\frac{2\epsilon}{\rho} e^{\rho T_k} \frac{(\rho T_k)^{r_k+1}}{(r_k+1)!} \leq e^{-q} \tag{EE40}$$

eventually. Indeed its logarithm is at most $a_0 q - 2q + O(q/\log \log q) + o(q) < -q$, since $a_0 < 1$. The same argument uses ρ_B and ϵ_B for D_B . The unperturbed observed exponential differs from I_B by at most cT_k . Intersecting the kernel of the rank- r_k perturbation with the variational subspaces for decreasing singular values gives

$$1 - \delta_k \leq s_j(U(t)) \leq 1 + \delta_k \quad (r_k < j \leq n_B - r_k), \quad \delta_k = cT_k + e^{-q}. \tag{EE41}$$

The upper bound follows from the codimension- r_k kernel; applying the same variational intersection for the lower bound loses r_k dimensions at the other end. If the displayed interval is empty the assertion is empty; on the actual rank bound it is nonempty eventually. At most $2r_k$ singular values are outside this band. EE38 bounds the absolute logarithm of every one by $O(q)$. Summing the first p decreasing logarithms therefore proves, uniformly for every exterior degree $1 \leq p \leq n_B$,

$$\sup_{N, 0 < |t| \leq T_k, 1 \leq p \leq n_B} \frac{|\log \|\wedge^p U(t)\||}{kq} = O_{h,\varpi} \left(\frac{1}{\log q \log \log q} \right) \rightarrow 0. \quad (\text{EE42})$$

The same holds intrinsically. This conclusion concerns the first-order multiplication evolution; the positive-energy word determinants retain their separate original maps and previously evaluated coefficients.

For the original terminal eigenclass $Mx = \omega x$, $\omega = k\gamma - ik\delta$, let $y = J^\dagger x$ and $\theta = \|y\|^2 / \|x\|^2$. On the exact RC35-44 terminal domain, $-\log \theta = O_{h,\varpi}(k \log q)$. The full identity

$$U(t)y - e^{it\omega}y = -J^\dagger(e^{itM} - I)\Pi x$$

gives

$$\frac{\|U(t)y - e^{it\omega}y\|}{\|y\|} \leq d_K(T_k, k) \sqrt{\frac{1-\theta}{\theta}} \leq e^{-\mathcal{L}_k + O_{h,\varpi}(k \log q)}. \quad (\text{EE43})$$

For completeness the intrinsic return has a separate proof. Put $A_j = e^\dagger C^j x$, $D_0 = |\omega| + c$. The eigenclass equation gives $\epsilon|A_0| \leq D_0\|x\|$, and $A_{j+1} = \omega A_j - g_j A_0$. Induction using $|\omega| \leq \rho$ and EE24 yields

$$\epsilon|A_j| \leq D_0(j+1)\rho^j\|x\|. \quad (\text{EE44})$$

EE26's vector comparison, paired now with x , implies

$$|\epsilon u^* C_B^j y - \epsilon A_j| \leq \epsilon(j+1)c^j \Xi_{N,k}\|x\| \quad (j \leq k).$$

Hence the free forcing satisfies, for $|t| \leq T$,

$$|\epsilon u^* e^{itC_B} y| \leq B(T)\|x\|,$$

$$B(T) = D_0(1 + \rho T)e^{\rho T} + \epsilon \Xi_{N,k}(1 + cT)e^{cT} + \epsilon e^{cT} \frac{(cT)^{k+1}}{(k+1)!}.$$

The exact scalar intrinsic Duhamel kernel has integral bound

$$\kappa_B(T) \leq \kappa_F(T) + \epsilon \Xi_{N,k} T e^{cT} + \epsilon_B T e^{cT} \frac{(cT)^{k+1}}{(k+2)!}. \quad (\text{EE45})$$

For the middle term integrate the finite series: $\int_0^T \sum_{j \geq 0} (j+1)(cs)^j / j! ds = T e^{cT}$. For the tail integrate the full factorial series. Thus no missing growing factor is hidden in EE45. At $T = T_k$, $\kappa_B = o(1)$ and $B(T_k) \leq e^{O(k \log q)}$. The scalar equation bounds its solution by $B\|x\|/(1 - \kappa_B)$. Its vector equation then gives

$$\|e^{itD_B} y - y\| \leq cT\|y\| + \frac{TB(T)}{1 - \kappa_B(T)}\|x\|.$$

Finally $|e^{it\omega} - 1| \leq T|\omega|e^{T|\omega|}$, so

$$\sup_{|t| \leq T_k} \frac{\|e^{itD_B} y - e^{it\omega} y\|}{\|y\|} \leq e^{-\mathcal{L}_k + O_{h,\varpi}(k \log q)}. \quad (\text{EE46})$$

This proves zero exponential transient gain for this same marked vector, while EE38 gives exponentially large maximizing gains. It does not permit differentiating a coarse error bound to assign the terminal current's sign. The actual imaginary interference term of TR20 remains.

8. Correct the constant collision and propagate it to RD15

The authoritative original kernel-volume coefficient is

$$C_{\partial} = 2J(1) = 1.354281987825292132\dots,$$

as fixed by 025 EP19 and its following paragraph, referring to 024 CK14. The distinct monic allowance difference is

$$C_{\text{monic}} = 2[\psi(0) - \psi(1)] = 0.350091046500433050\dots \quad (\text{EE47})$$

Their exact connecting map is RC10-12:

$$J' = 2\psi - 2(1+t)\psi', \quad C_{\partial} = 8 \int_0^1 \psi(t) dt - 8\psi(1) + 4\psi(0).$$

Thus there is an exact integral relation between them; they are not equal constants. For independent evaluation use $F(z) = 2K(\sqrt{z})/\pi$, $t = 4zF'(z)/F(z)$, $z = z_t$, and

$$J(t) = 2 \log F(z_t) - \frac{t}{2} \log z_t,$$

$$\psi(t) = (1+t) \log \frac{4}{\pi(1+t)(1-z_t)F(z_t)} + \frac{t}{4} \log z_t.$$

These retain B. C. Carlson's DLMF19.5 and 19.8.12 modulus conventions and are derived in the cited EP and RC proofs.

The published 027 RD15 used C_{∂} for C_{monic} . Its exact denominator inequalities remain valid. Its corrected four-cutoff consequence is

$$\boxed{\begin{aligned} \mathcal{R} \log \mathfrak{A}_{\lambda, N} &= 4[\psi(0) - \psi(1)]q + O_{h, \varpi}(k \log q) \\ &= 0.700182093000866100\dots q + O_{h, \varpi}(k \log q). \end{aligned}} \quad (\text{EE48})$$

Indeed RD's separate denominator calculation gives $\log \mathfrak{A}_{\lambda, N} = 2q\psi(s_N) + O(k \log q)$; apply the four endpoint estimates used in EE39. The corrected scalar equals the lower propagator plateau, with the exact maps for each quantity supplied above. The kernel coefficient remains C_{∂} throughout.

9. Verification and scope

The accompanying checker verifies the determinant lemma, original/full and observed rank-one algebra, exact Taylor coefficients, measured inverse defect, its determinant sign, the jet-fibre comparison on finite positive metrics, feedback-coefficient bounds, the sharp shear singular-value formula and independent scalar profile values. It includes controls that fail if observation is treated as multiplicative, if time reversal is substituted for the measured inverse, or if the two coefficient names are identified.

Finite examples supplement the analytic proofs and are identified as auxiliary matrices. They are not evaluations of an actual off-line zero or period. This derivation uses the complete previously proved source comparisons and uniform monic theorem, with exact locators above. It does not claim a fresh audit of every proof retained in the cumulative source bank. The incoming formulas survive after the explicit repairs EE4, EE8, EE34-35 and EE48; the growing-jet step is fully established by EE9-21.